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Palaeomagnetism of the (Late Mesoproterozoic) Stoer Group, northwest Scotland: implications for diagenesis, age and relationship to the Grenville Orogeny

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Abstract – The Stoer is the lowest of three groups comprising the Torridonian Supergroup, a clastic succession deposited late in Mesoproterozoic times on the (Lewisian) metamorphic foreland to the Caledonian Orogen in northwest Scotland. This study reports a palaeomagnetic, rock magnetic and magnetic fabric investigation through the full stratigraphic thickness of the succession. A primary magnetic fabric of sedimentary origin defines current flow from a westerly veering to northwesterly source. Rock magnetic studies identify the presence of both magnetite and hematite in these sediments. Magnetite is apparently of primary detrital origin whilst the hematite probably results mostly from early diagenesis in an environment of restricted chemical weathering. Palaeomagnetic study of sedimentary slumps shows that magnetic remanence post-dates deposition but was probably fixed by early dewatering and lithification because slumped blocks of Stoer in basal Torridon Group sediments preserve a primary remanence. Tilt adjustment, although inconclusive, also implies that magnetic remanence is older than pre-Torridian Group tectonic deformation. The lower part of the Stoer succession shows a progressive increase of magnetic inclination with shallower components resident in magnetite and steeper components in hematite. The succession above the Stac Fada Member has the steepest magnetic inclination and shows no significant difference between magnetite and hematite component directions. The inferred time sequence of palaeopoles coincides with the Gardar palaeomagnetic track ($\sim$ 1250–1130 Ma) at 1180 Ma, conforming to a Pb–Pb determination of 1199 ± 70 Ma. The Stoer Group was fully lithified and deformed before deposition of the Torridon Group at $\sim$ 1030 Ma because it contains no vestige of the range of Caledonian and later overprints found extensively in the latter. Sedimentation and lithification of the Stoer Group are therefore linked with a phase of extensional tectonism at 1200–1150 Ma and deformation is attributed to a culminating phase of deformation in the nearby Grenville Belt at $\sim$ 1100 Ma.

Keywords: Proterozoic, palaeomagnetism, Scotland, Torridonian Supergroup, Stoer Group, diagenesis, age.

1. Introduction

The Stoer Group is the oldest of three successions comprising the Late Mesoproterozoic Torridonian Supergroup in northwest Scotland and is composed predominantly of arkosic red sandstones. It is interpreted as a rift-controlled clastic suite of fluviatile and lacustrine origin linked to syn-depositional faulting, but also includes a few thin limestones hosting Riphean age microplankton (Cloud & Germs, 1971). Palaeocurrent reversals in lake deposits indicate a rift origin between active marginal faults (Stewart, 1982) with a provenance in the local underlying Lewisian basement (Rainbird, Hamilton & Young, 2001) and deposition occurring within enclosed basins in an arid setting (Stewart, 1991). An unconformity between the Stoer and the succeeding Sleat and Torridon groups was first inferred from the contrast in their palaeomagnetic records (Irving & Runcorn, 1957) and subsequently confirmed by field evidence (Gracie & Stewart, 1967).

Whole rock Rb–Sr isochron determinations of 991 ± 24 Ma and 796 ± 24 Ma on mudrocks by Moorbath (1969) originally indicated a Neoproterozoic age and a considerable time gap between deposition of the Stoer and Torridon groups, as also implied by the angular unconformity between them and their contrasting magnetization records. However, it was also observed that the magnetizations, shown to be of quasi-primary origin, yield pole positions correlating with the Laurentian apparent polar wander path (APWP) at $\sim$ 1100 and $\sim$ 1000 Ma (Smith, Stearn & Piper, 1983); no agreement was found with poles at the times defined by ages of Moorbath (1969). The need for refined age dating has been confirmed by Turnbull, Whitehouse & Moorbath (1996), who derive a Pb–Pb age of 1199 ± 70 Ma from limestone in the Stoer Group and a Rb–Sr isochron of 997 ± 39 Ma from mudrocks of the Torridon Group. Downie (1962) identified

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Table 1. Summary of previous palaeomagnetic results from the Stoer Group, northwest Scotland

Study		D/I°	N	K	α_{95}	Pole °N	Pole °E	dp/dm°
Irving & Runcorn (1957)	Stoer Group	307/34	13*	40	7	35	248	5/8
Stewart & Irving (1974)	SFM, Stoer	313/29	20*	18	8	35	234	5/9
	SFM, Enard	308/37	3*					
		283/53	12*					
Smith, Stearn & Piper (1983)	Stoer	315/27	16	17	9.2	35	231	5/10
	Enard	293/49	7	37	10.1	37	263	9/13
Piper & Poppleton (1991)	Stoer	312/21	9*	31	9.4	31	233	5.2/9.8
	Enard	303/20	7*	58	8	26	242	4.4/8.4
Torsvik & Sturt (1987)	Stoer	318/16	7	42.4	8.1	31	225	4/8
	Achiltibuie	315/17	8	40.4	7.8	30	228	4/8
	Overprint	30/52	33*	21.1	6.1	58	123	6/8

D and I are the magnetic declination and inclination derived from N sites (* = samples); k is the Fisher precision parameter and α_{95} is the radius of the cone of 95 % confidence about the mean direction of magnetization. dp and dm are the radii of the oval of 95 % confidence about the pole position in the colatitude direction and at right angles to it, respectively. SFM = Stac Fada Member.

nannofossils, probably marine phytoplankton, within limestone and red shales from Stoer attributed to a Middle Riphean age (1350–1050 Ma) and compatible with evidence of Cloud & Germs (1971). Uplift and cooling ages as young as ~ 1100 Ma on the underlying Lewisian metamorphic basement (Cliff & Rex, 1989) imply that rifting and sedimentation occurred shortly after final uplift of the basement on which the Stoer lies with topographic unconformity. Subsequently ~ 2 km of fluviatile and lacustrine sedimentation was controlled by syndepositional fault displacements (Nicholson *in* Anderton, Gibbons & Nicholson, 1992).

Stoer Group rocks were included in an early palaeomagnetic study by Irving & Runcorn (1957) that resolved a single polarity NW+ Natural Remanent Magnetization (NRM, see Table 1). This contrasted with a dual polarity NW−/SE+ magnetization recovered in the Torridon Group and led the authors to conclude that an unconformity representing an appreciable time interval was present between the two groups. Stewart & Irving (1974) repeated some of the sampling of Irving & Runcorn (1957), this time conducting partial thermal demagnetization to 500 °C, and resolved comparable directions to the earlier study. They were able to confirm that the change in remanence direction coincided with the unconformity, which had by then been demonstrated by geological mapping (Stewart, 1966). The probable remanence carrier in the sediments was identified as specular hematite, although magnetite was identified as the carrier in the Stac Fada Member, a volcanogenic unit near the middle of the succession. A regional palaeomagentic study of the Torridonian Supergroup by Smith, Stearn & Piper (1983) confirmed a uniform magnetic polarity in the Stoer Group but found a significant difference in magnetic inclination between outcrops in the south (Toscaig, I = 51°) and in the type section at Stoer (I = 29°). Both magnetite and hematite were recognized as remanence carriers. Torsvik & Sturt (1987) conducted further studies with comprehensive demagnetization and field tests; a shallower remanence inclination than that recognized in most previous studies suggested that the latter might have incompletely removed a small present field component (Table 1).

The present investigation of the Stoer Group has four objectives: (1) to resolve components of magnetization through the full stratigraphic thickness of the Stoer Group by comprehensive thermal demagnetization; (2) to undertake slump and conglomerate tests to help constrain the time at which the remanence was acquired; (3) to improve age constraints because the geochronological evidence has relatively large error limits (± 70 Ma), and hence clarify the tectonic setting; (4) to undertake rock magnetic and magnetic fabric studies to aid in interpretation of magnetic carriers and sediment transport directions.

2. Geological framework

The Torridonian Supergroup was deposited onto a deeply eroded metamorphic basement comprising the Lewisian Complex. This basement has an origin in Late Archaean and Early Palaeoproterozoic crust-forming events, of which the most important was the Scourian Episode at ~ 2500 Ma, and it was subjected to regional reworking in the north and south during the Laxfordian Episode at ~ 1800 Ma. The mainland and Hebridian outcrops of the basement comprise the Hebridean Craton (Watson, 1975). They are offset by ~ 97 km of sinistral motion along a fault, now located within the Minches Basin, late in the Caledonian orogenic cycle and relative rotational uplift to the west along low angle fault(s) has caused the Torridonian cover to be removed by erosion from the Hebridean sector (Piper, 1992).

The Hebridean Craton comprised the northwest foreland to the Caledonian Belt during early Palaeozoic orogenesis and is identified as a former peripheral part of the Laurentian Craton of North America and Greenland during its long Precambrian history (Fig. 1). Large areas of this shield were reworked by



Figure 1. The Laurentian Craton in Proterozoic times, showing the major divisions of tectonothermal terranes including the Grenville Belt. Also shown are zones of Mesoproterozoic rift development and the location of the Hebridean Craton including the area of the present study. GAD is the Great Abitibi Dyke.

tectono-thermal events between ~1900 and 1700 Ma in the 'Hudsonian' episodes which include the Laxfordian in northwest Scotland. Relict nuclei such as the Scourian escaped reworking and comprise the Archaean cratons within the shield. Later in Mesoproterozoic times, by which time they had been uplifted and cooled, the metamorphic terranes were subjected to brittle fracture in tensile stress regimes responsible for dyke emplacement and formation of rift systems incorporating sedimentary–volcanic successions. These are well represented along the southeastern margin of the shield (Fig. 1) and include the Gardar Rift in South Greenland and the Mid-Continental rift, which incorporates the Keweenawan Province, in central North America. The last Proterozoic mobile event to affect the Laurentian Shield was the Grenville Episode which developed along an Andean-type margin (Hanmer *et al*. 2000). Although tectonism culminated at ~1100 Ma, some parts of this tectonic province have a history extending back at least ~300 Ma prior to these times.

Preservation of the Stoer Group is limited to a number of isolated and tilted outcrops (dips $< 30^{\circ}$) beneath the Torridon Group (Fig. 2). These are now exposed between Stoer in the north and Loch Maree in the south (Fig. 2) where they lie entirely on Scourian (~2500 Ma) metamorphic terrane. They are composed predominantly of sandstones, conglomerates and mudstones with the clastic facies considered to be locally derived sediments deposited as fans in fluviatile and lacustrine environments (Stewart, 1975) constrained to a fault-bounded basin (Van de Kamp & Leake, 1997). Stewart (1982) divides them into marginal and exotic facies. The marginal facies include locally derived gneiss breccia grading laterally into red alluvial sandstones and shales; a thin limestone horizon is also found in this facies (Stewart, 1990). The exotic facies consists of cross-bedded sandstones containing rounded gneiss and quartzite pebbles; some of these are typical of underlying Lewisian rock types.

Rainbird, Hamilton & Young (2001) report concordant Pb–Pb ages from four sandstone units of the Stoer Group. Ages are concentrated in the range 2930–2480 Ma, consistent with the view that the sandstone detritus was derived from local basement of the Scourian Complex. A close chemical affinity with the underlying basement is also demonstrated by major element analyses and Th/Sc ratios (Young, 1999); the latter are consistent with Archaean basement. Enrichment of certain elements, notably Ca, in the Stoer Group has been taken to suggest that volcanic ash may have contributed to the Stoer sedimentation (Stewart, 1990). However, Young (1999) shows that major and minor element ratios in the argillaceous facies have chemical characteristics typical of regimes of inhibited chemical weathering. A dominance of physical weathering could be a signature of periglacial environments, and glaciation has been invoked to explain the origin of conglomerates and breccias in the Stoer Group (Davison & Hambrey, 1996). However, a Proterozoic arid environment is more likely (Young, 1999); this is supported by the low palaeolatitudes of deposition (equivalent to 10–33° from the values listed in Table 1) although such evidence is not unambiguous in Proterozoic times (Williams, 1998). In support of this view, Young (1999) has re-examined the supposed diamictites in the Stoer Group and found that they pass laterally into mudstones showing dessication cracks and ripple marks. He concludes that the supposed dropstones of Davison & Hambrey (1996) are boulders isolated by rolling or sliding ahead of subaerial fans into shallow ephemeral lake deposits.

3. Field sampling

The major outcrop of the Stoer Group is on the Stoer Peninsula (Fig. 3a), where the succession commences with locally derived breccias and conglomerates succeeded by laminated sandstones and mudstones of lacustrine origin. These beds are followed by trough cross-stratified red sandstones, probably mostly of fluviatile origin (Stewart *in* Barber *et al*. 1978);

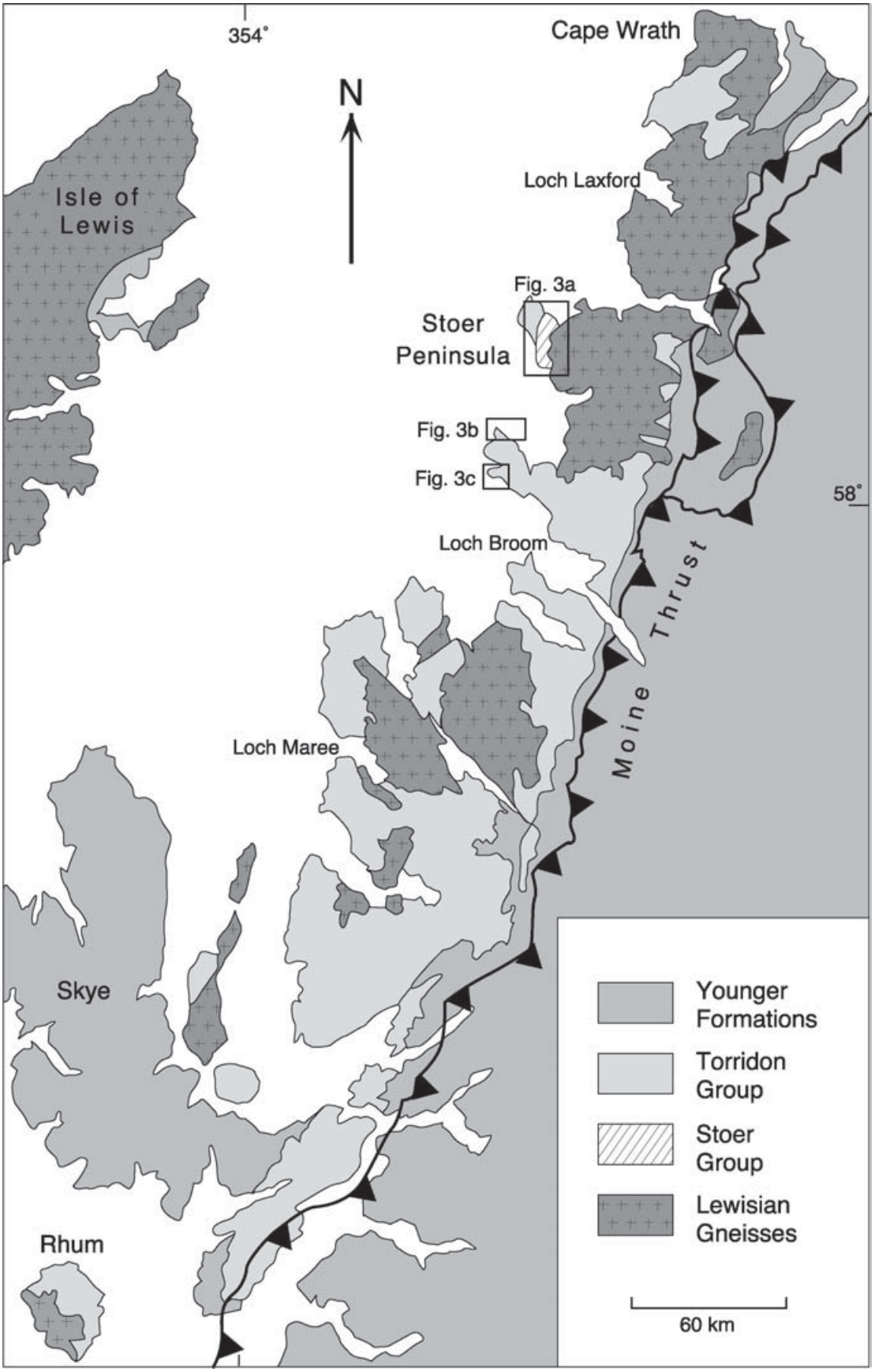


Figure 2. Outline geological map of the mainland outcrop of the Hebridean Craton of northwest Scotland, showing the distribution of the Torridonian Supergroup and the Stoer Group outcrops sampled for this study (enclosed by boxes and expanded in Fig. 3).

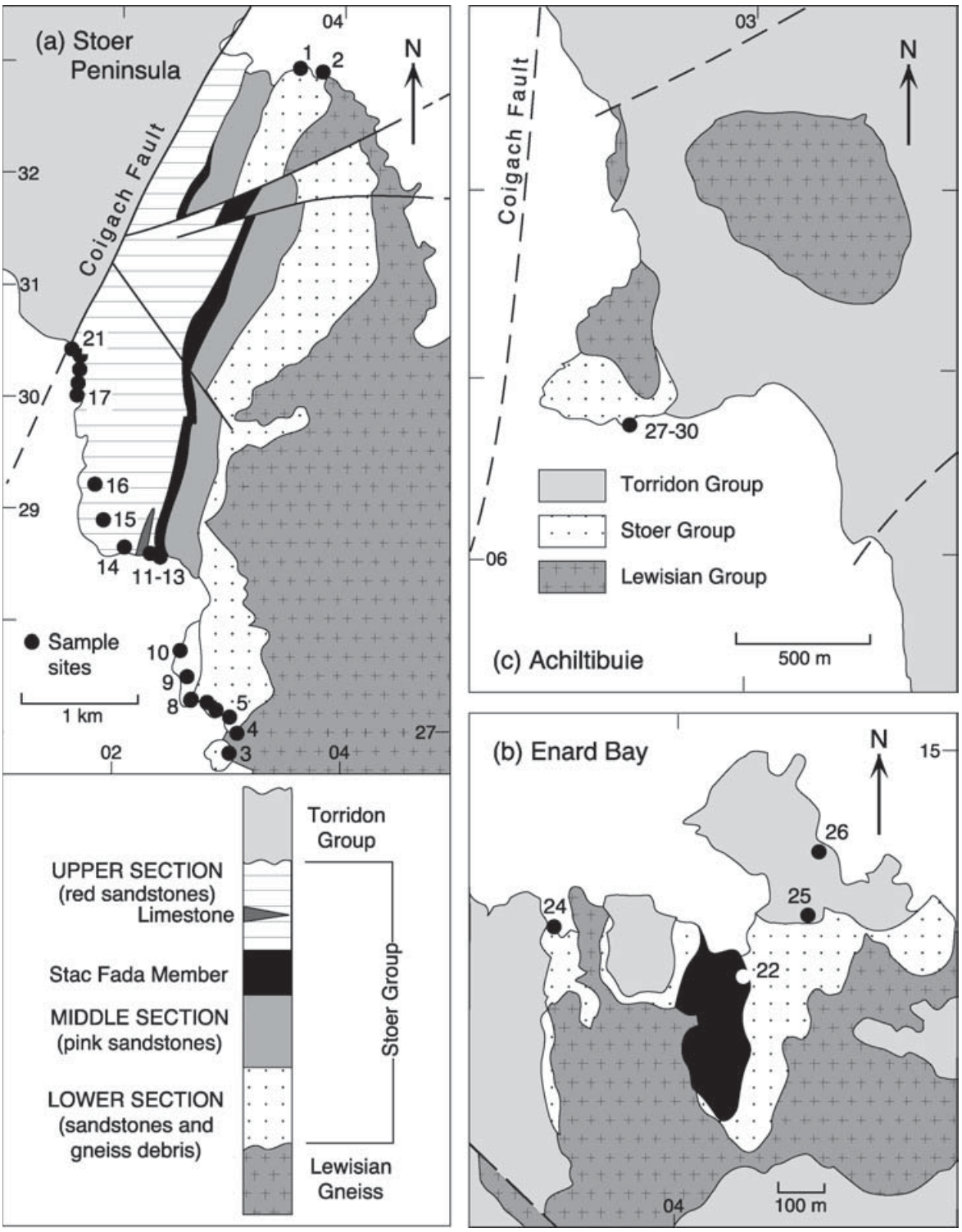


Figure 3. Local geological maps after Stewart *in* Barber *et al*. (1978) of the three sample locations with palaeomagnetic sites. The regional location of these areas within the outcrop of the Torridonian Supergroup is shown in Figure 2.

sampling sites 3–10 are within these beds and include penecontemporaneous slumps at sites 6 and 10. The Stac Fada Member (site 11, lower part, site 12 upper part) is a 12 m thick unit interpreted either as a volcanic mudflow (Lawson, 1972) or an extrusion of fluidized pepperite (Sanders & Johnston, 1989). Site 13 includes sandstone blocks incorporated into the Stac Fada Member and site 14 is a red mudstone resting on this member. Succeeding sites (14–21) are planar and cross-stratified sandstones that extend the sampled sequence to its exposed limit where it is downfaulted against the Torridon Group rocks along

the Coigach Fault (Fig. 3a). On the north side of the peninsula, site 1 is in metamorphic clasts and sandstone matrix at the unconformity with the Lewisian complex and site 2 is in a breccia bed at Reinachait. These latter sites were collected to conduct palaeomagnetic conglomerate tests and specifically to test for any regional magnetic overprinting within the Stoer Group.

The thin Stoer Group succession exposed on the southern shore of Enard Bay is overlain unconformably by the Diabaig and Applecross formations of the Torridon Supergroup (Fig. 3b). The Stac Fada Member (sites 22 and 23) is overlain by red sandstones (site 24) and conglomerates. Sites in immediately overlying beds are in the Diabaig (site 25) and Applecross (site 26) formations. South of Achiltibuie (Fig. 3c), gneiss breccia overlain by flaggy red sandstones mantles Lewisian topography and blocks of bedded and lithified Stoer sandstone have slipped down a palaeoslope to become incorporated into sandstones of the Diabaig Formation during deposition of the latter (Stewart *in* Barber *et al*. 1978). Multiple cores were drilled from large blocks of the Stoer Group here for a conglomerate test (site 27). Samples are listed with coordinates in Table 2 and approximate stratigraphic locations are given in Figure 4.

Table 2. Summary of palaeomagnetic sampling sites in the Stoer Group of the Stoer Peninsula

Stoer Group (58.2N, 354.5E)	Site no.	No. of cores	Grid reference	Regional tilt °Dip/°E
Red arkosic sandstones	21	6	25118305111	20/270
	20	7	25118305111	20/270
	19	13	25118305111	20/270
	18	7	25118305111	20/270
	17	7	25118305111	20/270
	16	7	25150285031	25/270
	15	7	25195285073	26/270
Red mudstone	14	22	35036285080	24/270
Stoer blocks in SFM	13	3	35062285108	26/270
Upper part of SFM	12	5	35062285108	26/270
Lower part of SFM	11	9	35062285108	26/270
Arkosic red sandstones	10	7	35125275029	27/270
	9	7	35114275159	27/270
Slump in sandstone	8	6	35153275161	27/270
Sandstone	7	5	35153275161	27/270
Slump in sandstone	6	8	35153275161	27/270
Arkosic red sandstones	5	9	35153275161	27/270
	4	7	35196265014	27/270
	3	6	45005265004	27/270
Lewisian clasts	2	13	45195235016	31/290
	1	21	45149325010	31/290

SFM – Stac Fada Member.

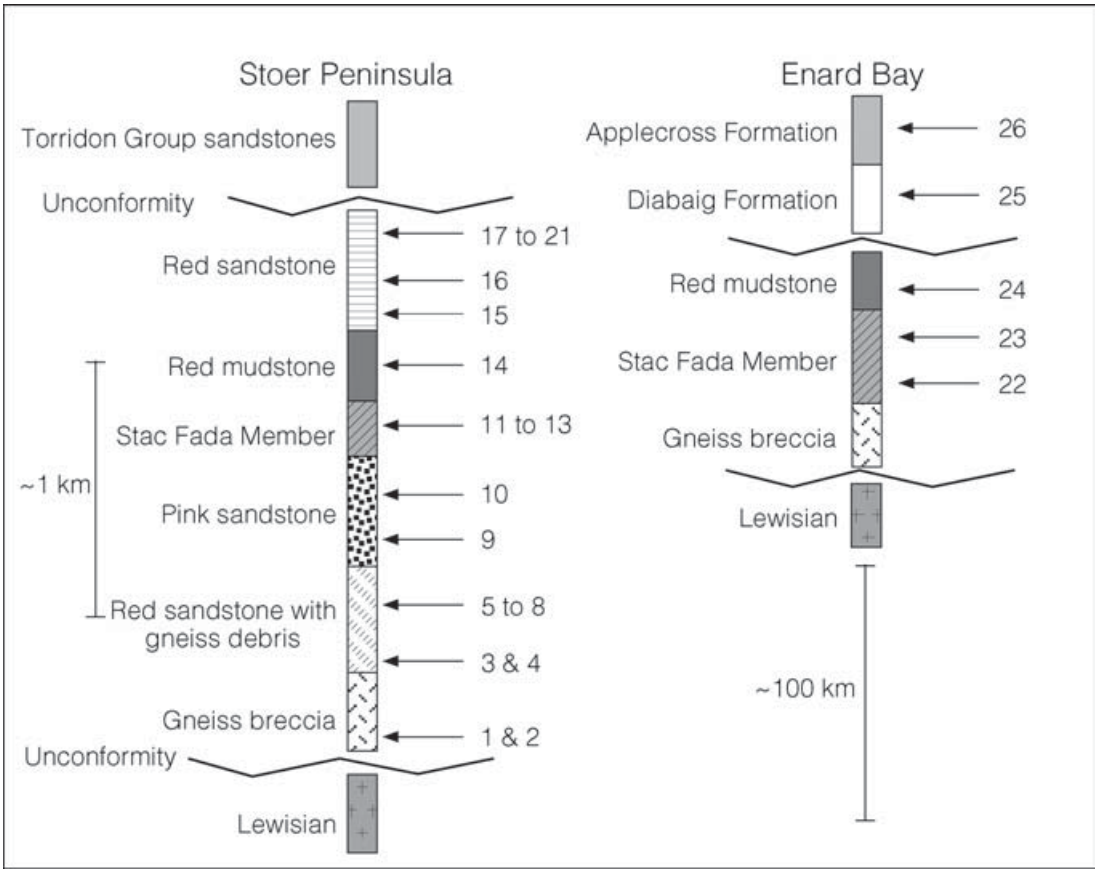


Figure 4. Simplified stratigraphic successions through the Stoer Group showing approximate levels of the palaeomagnetic sites.

All cores were drilled in the field using a portable gasoline powered drill and oriented by sun and magnetic sights prior to extraction. Cylinders from each sampled core were measured using a nitrogen SQUID magnetometer and NRMs were subjected to progressive thermal demagnetization in steps of 100 °C reduced to 50° and 20 °C as the Curie points of the magnetic carriers were approached. The results of this treatment were evaluated as orthogonal projections of the remanence vector at each stage of treatment; components of magnetization were recognized by eye and their directions determined by principal component analysis. Cuttings from the field cores were used for a range of rock magnetic tests as described in the next section.

4. Rock magnetic results

4.a. Frequency-dependent susceptibility ($\kappa_{fd\%}$)

Susceptibility measurements using an alternating magnetic field at two different frequencies (0.46 and 4.6 kHz) are sensitive to the single domain–superparamagnetic (SD–SP) boundary in magnetite ($\sim$ 30 nm). The difference in susceptibility value is usually expressed as a percentage, that is, $\kappa_{fd\%} = ((\kappa_{lf} - \kappa_{hf})/\kappa_{lf})*100$ in which κ_{lf} and κ_{hf} are the low and high frequency susceptibilities, respectively. Values of $\kappa_{fd\%} < 5$ % occur when SD–MD (single domain–multi-domain) grains are dominant, whereas values > 5 % are expected for materials containing significant proportions of SP grains (Dearing *et al*. 1997). When lower susceptibilities are observed at higher frequency they indicate the presence of SP grains because these grains oscillate at low frequencies but are unable to follow high frequency changes and keep in phase with the field.

Theoretical analysis shows that grains < 30 nm in size have contrasting susceptibilities at frequencies of 0.45 and 4.5 kHz (Dearing *et al*. 1996), whereas grains > 30 nm in size show no difference. The value of $\kappa_{fd\%}$ increases non-linearly up to 15 nm and then decreases up to 30 nm with theory predicting a maximum value of 14–17 in the value of $\kappa_{fd\%}$ for grains of 15 nm in diameter. Figure 5 shows the variations of $\kappa_{fd\%}$ as a function of stratigraphic height through the Stoer Group succession. Values of $\kappa_{fd\%}$ are mostly about 5 %, implying mixed grain sizes but including small grains in the SP range. In the section of the Stoer Group above the level of the Stac Fada Member, $\kappa_{fd\%}$ increases roughly linearly with stratigraphic height, meaning that finer ferromagnetic grain sizes become more important towards the top of the succession. This is probably because these sediments higher in the section have travelled further, resulting in more sorting, and consequently loss of the heavy mineral fraction.

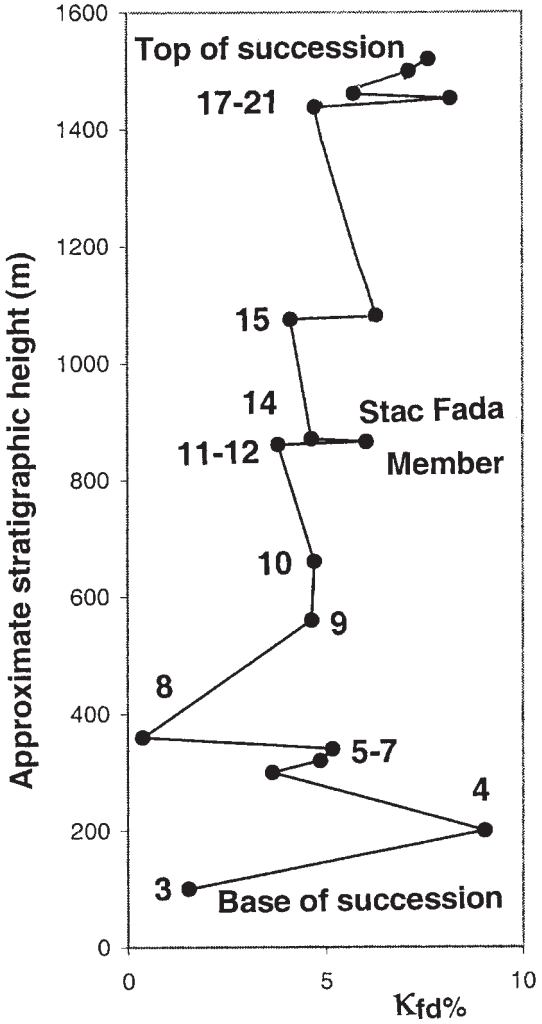


Figure 5. Variation of the low-high susceptibility difference parameter, $\kappa_{fd\%}$, through the Stoer Group succession in the type area at Stoer Bay.

4.b. Magnetic fabrics

Anisotropy of magnetic susceptibility (AMS) can be a significant parameter in sedimentary rocks when grains are aligned by current flow during deposition or grow under hydrostatic or tectonic stress fields at later stages of diagenesis. The variation of susceptibility with direction is represented by a 3-3 symmetric tensor and is approximated by a triaxial ellipsoid. The principal directions of the ellipsoid are denoted k_1 or $k_{max} > k_2$ or $k_{int} > k_3$ or k_{min}. The degree of anisotropy (Pj) is defined by k_{max}/k_{min}; lineation (L) is defined by k_{max}/k_{int} and foliation (F) by k_{int}/k_{min}. When $F > L$ the ellipsoid is oblate and when $L > F$ the ellipsoid is prolate in form. The AMS ellipsoid is usefully analysed using the shape parameter, T (Hrouda, 1982) where:

$$T = (2(\ln k_2 - \ln k_3)/(\ln k_1 - \ln k_3)) - 1$$

When $-1 < T < 0$ the ellipsoid is prolate and when $0 < T < 1$ the ellipsoid is oblate; T is zero when the ellipsoid is neutral.

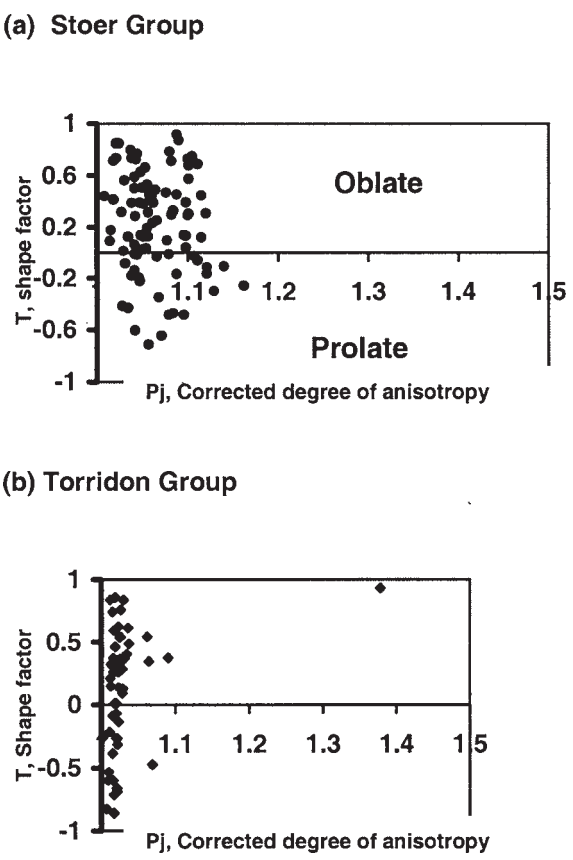


Figure 6. (a) Comparative distributions of the shape parameter, T, of the anisotropy of magnetic susceptibility (AMS) plotted as a function of the degree of anisotropy (Pj) in the Stoer Group; also shown for comparison (b) are results from two formations (Aultbea and Diabaig) in the Torridon Group.

AMS has been determined in selected cores from most sites in the Stoer Group using a low field KLY-3 Kappabridge. The majority of samples have less than 10 % anisotropy and show a predominance of oblate fabrics consistent with an origin by sedimentation (Tarling & Hrouda, 1993). A comparison with two formations of the overlying Torridon Group, however, shows that the Stoer Group is significantly more anisotropic than the Aultbea and Diabaig formations (Fig. 6). In both cases a small percentage of ferromagnetic grains are likely to dominate over the influence of the diamagnetic silicate (quartz and feldspar) matrix and the contrast is probably due to relative contributions of primary grains aligned by flow. In the Stoer Group, magnetic remanence proves to be a post-depositional remanent magnetization (PDRM, Section 5), whereas all formations of the Torridon Group show appreciable degrees of magnetic overprinting as a chemical remanent magnetization (CRM) attributed to diagenetic precipitation of ferromagnetic grains unrelated to primary deposition.

The directions of principal axes of susceptibility adjusted for tilt of the beds are shown in Figure 7. k_1 and k_2 are found to be distributed in planes inclined by 20–30° to the bedding whilst k_3 is perpendicular to this plane and displaced 20–30° from the vertical. This distribution is similar to examples illustrated by laboratory sedimentation of sands (Rees, 1961) and defines a primary imbrication due to stacking of cross-bedding. The direction of k_1 for the sediments below the Stac Fada Member is E to ENE consistent with a current flow direction from the west (Fig. 7a). In the beds above this member, the general directions are from W to WSW (Fig. 7b). Most authors consider that the source of the Torridonian sediments was formerly sited to the west of the present outcrop (Nicholson, 1993; Rogers *et al*. 1990; Allen, 1991; Van de Kamp & Leake, 1997; Rainbird, Hamilton & Young, 2001), although Stewart (1982) considers that easterly derivation also occurred in the Stoer region.

The fabric in the Stac Fada Member at Stoer is similar to the sediments above with a k_1–k_2 plane inclined to the SE at about 30° and k_3 dipping steeply NW perpendicular to this plane (Fig. 7c). At Enard Bay the k_1–k_2 plane in the same unit dips more shallowly at about 15° to the south (Fig. 7d). This fabric also seems to be primary but is developed in a unit that was deposited quickly by an event linked to volcanic activity (Stewart, 1990; Sanders & Johnston, 1989). If the fabric was acquired by an advancing mudflow it could be explained as an imbricate effect as the mudflow moved from the NW direction at Stoer and from the NNW at Enard Bay (see also Young, 2000). The lower inclination of the fabric at Enard Bay could then be due to a greater distance from the source area and perhaps to a more planar flow.

4.c. Temperature variation of magnetic susceptibility and saturation remanence

During progressive heating as part of the thermal demagnetization procedure the susceptibility of sediment samples drops slightly (Fig. 8a). This probably reflects the contribution of paramagnetic minerals obeying the Curie Law with $k \propto 1/T$ where T is absolute temperature. Above $\sim 500\,^{\circ}C$ there are variable increases in susceptibility which are larger in the lower part of the succession with a higher argillaceous content. This is explained by a reconstitution of the clay minerals during the heating. It does not appear to influence the small ferromagnetic content because magnetic component behaviour is unaffected. Below room temperature, magnetic susceptibility continues to increase in most samples although a suppressed Morin transition in hematite below $-15\,^{\circ}C$ is also usually detected. A minority of samples also show a Verwey transition at $-150\,^{\circ}C$. This temperature is the isotropic point of MD magnetite where magnetocrystalline energy $\rightarrow 0$ and the grain changes from being an electric insulator to a semi-conductor (Fig. 8b).

The presence of hematite is dominant in thermomagnetic measurements (saturation magnetization, M_{sat},

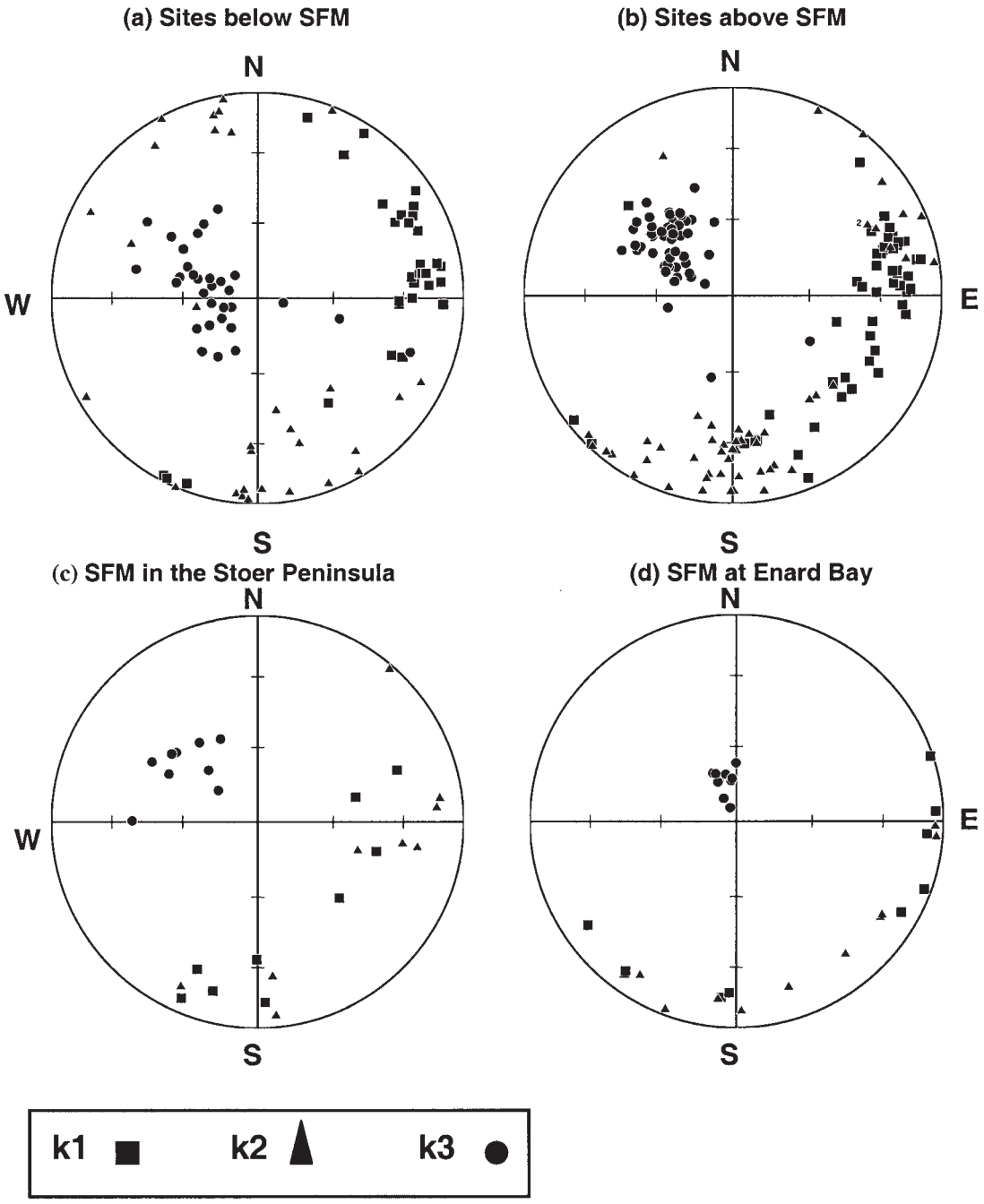


Figure 7. Sample level directions of k_1, k_2 and k_3 for Stoer Group sediments and the Stac Fada Member (SFM) adjusted for tilt of bedding. Lower hemisphere equal area projections.

versus temperature, T) determined here using a horizontal translation automated Curie balance (Fig. 8c). Since hematite has a saturation remanence only about 2 % of that of magnetite it needs to be present in substantially greater amounts than the magnetite to be detected. In the case of the Stoer Group sediments the Curie temperature of low-Ti hematite (~ 680°) is dominant, although an inflection is sometimes present at the Curie point of magnetite (~ 580 °C). The presence of small amounts of magnetite is also evident from a slight reduction in M_s on cooling (Fig. 8c); this is always present and attributable to the oxidation of magnetite (with high M_s) to hematite (with lower M_s) in the oven. The thermomagnetic experiments therefore show that hematite is the dominant ferromagnetic mineral in the Stoer Group sediments although small amounts of magnetite, presumably as detrital grains derived from metamorphic rocks of the underlying Scourian basement, are always present as well.

4.d. Hysteresis and IRM study

Hysteresis behaviour is determined by exposing the rock to an increasing applied field (H) and determining

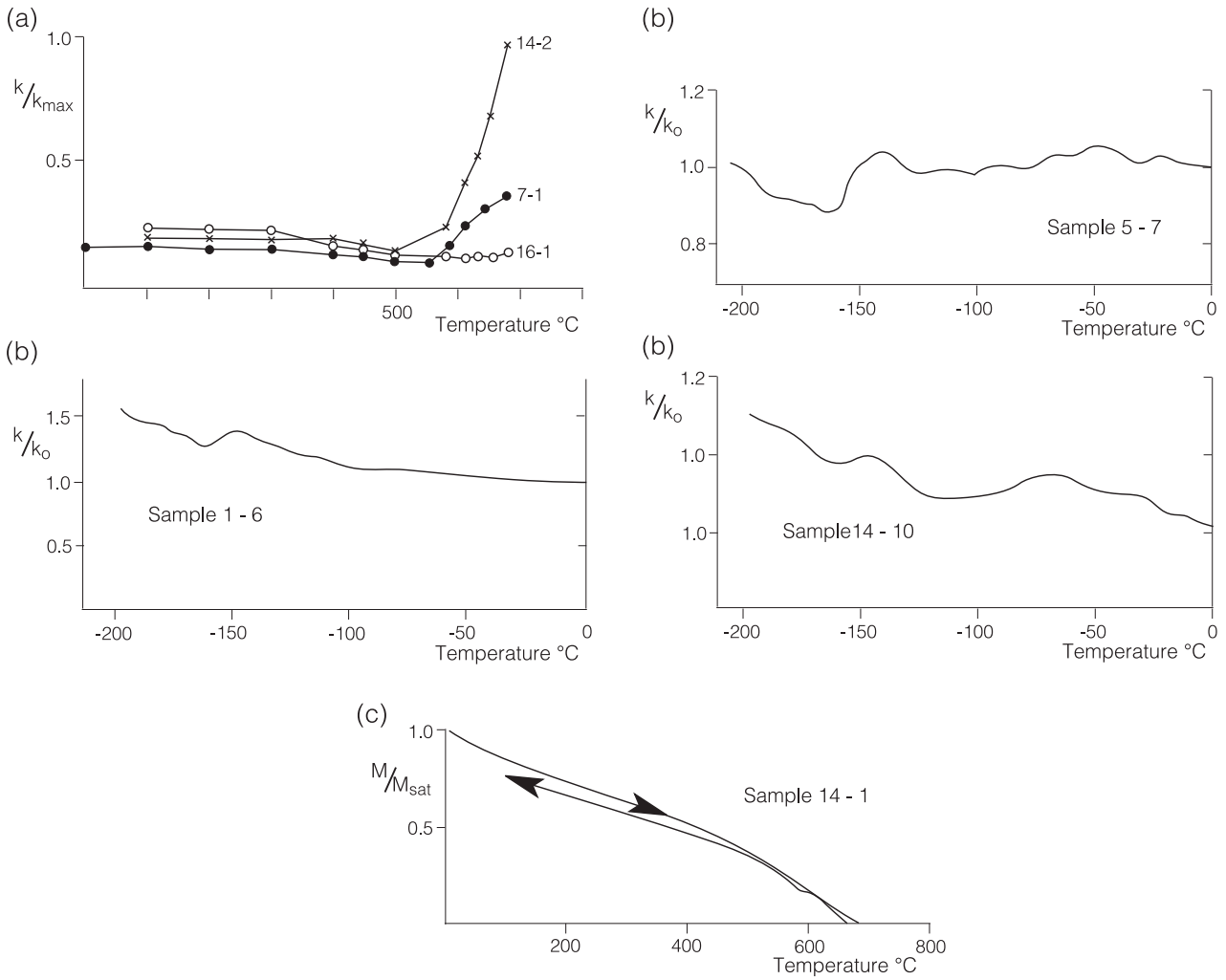


Figure 8. Typical examples of variation of magnetic behaviours in Stoer Group sediments with changes in temperature. (a) Variation of magnetic susceptibility with increasing temperature during demagnetization. (b) Three types of variation of magnetic susceptibility (see text) with decreasing temperature. (c) Thermomagnetic determination showing the change of saturation magnetization (M_{sat}) with increasing temperature; the arrows distinguish heating and cooling curves.

the magnetization acquired. When saturation (M_s) is achieved at H_{max}, a progressively increasing backfield is applied to subtract the acquired field. The additional field when the backfield has reached $-H_{max}$ needed to reduce the residual magnetization (M_{rs}) to zero is the coercivity (H_c). By cycling the applied field, a full hysteresis loop is obtained with rapid saturation indicative of low coercivity minerals such as magnetite and slow saturation indicative of high coercivity minerals such as hematite. Constricted or wasp-waisted loops result when a mixture of different magnetic minerals of different grain sizes is present (Roberts, Cui & Verosub, 1995). A linear component to the loop with $H \propto M$ is the signature of paramagnetism.

Hysteresis loops determined for samples of the Stoer Group sediment using a variable field translation balance are always wasp-waisted, showing that a mixture of ferromagnetic components is present in these rocks (Fig. 9a). A linear extension is the signature of paramagnetism; this is probably mostly resident in iron-bearing clay minerals and fine hematite pigment.

The variable field translation balance has a maximum magnetizing field of 0.8 Tesla (T) which is insufficient to fully saturate most hematite grain sizes, and the interpretation of the extended parts of the hysteresis loops is ambiguous unless it is clearly linear and due to paramagnetism. Accordingly, Isothermal Remanent Magnetizations (IRM) have been progressively imparted to samples using a pulse magnetizer in fields up to 2.7 T. Saturation is largely but not completely achieved in this maximum field, giving a clear signature that magnetite is present, although the curves always have an inflection showing that both low and high coercivity components are present in these rocks (Fig. 9b). Robertson & France (1994) show that acquisition of IRM can discriminate between multiple components if the magnetization gradient is plotted against the log of the applied field. The curve shapes isolate

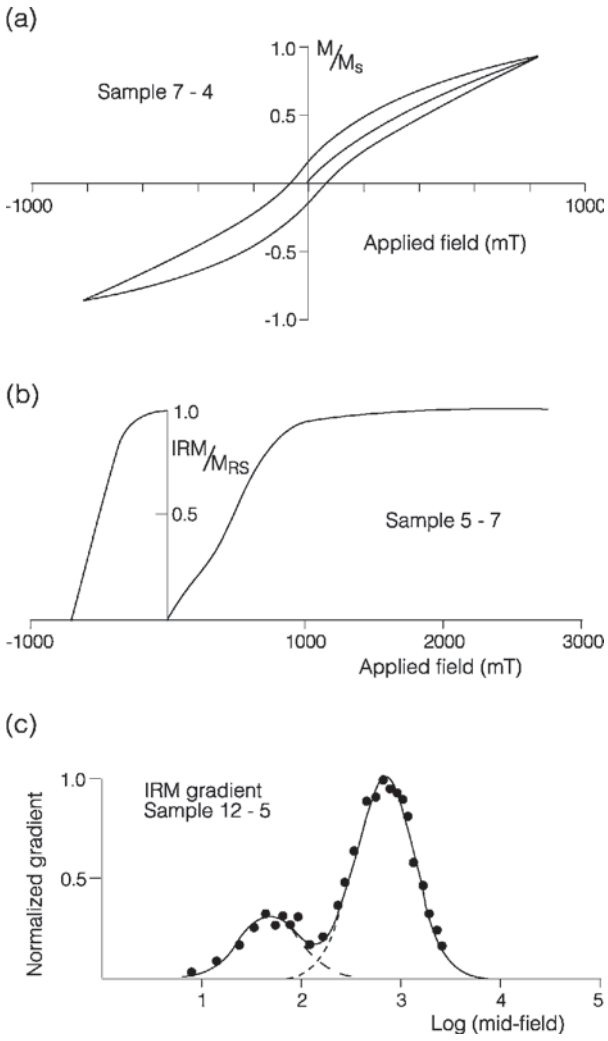


Figure 9. Typical behaviours of Stoer Group sediments in applied magnetic fields. (a) Hysteresis curve determined by variable field translation balance to a maximum applied field of 0.8 Tesla. (b) Isothermal Remanent Magnetization (IRM) determination to maximum field of 2.7 Tesla. (c) Magnetization gradient from an IRM determination plotted as function of the log of the midfield (Robertson & France, 1994). See text for explanation.

the typical coercivities and the relative importance of each ferromagnetic component. Figure 9c is a typical example from the Stoer Group and shows that a low coercivity component (typically with $H_{cr} = 0.2$–0.5 T) is mixed with a much larger amount of high coercivity component (typically $H_{cr} = 0.8$–1.8 T). Thus the analyses of hysteresis and IRM both endorse the thermomagnetic analyses by showing that hematite (of pigmentary, detrital specularite and/or authigenic origin) is the dominant ferromagnetic constituent in the Stoer Group, but they also show that variable small amounts of detrital magnetite are also present. This latter observation implies that oxidation of magnetite to hematite has not gone to completion in these rocks and supports the observation, based on geochemistry, that chemical weathering appears to have been inhibited during deposition of the Stoer Group (Young, 1999).

5. Palaeomagnetic results

5.a. The Stoer sedimentary succession

Most of the sediment magnetizations show a two-component structure with the components residing broadly in 'magnetite' and 'hematite' spectra resolved between 0 and 580 °C, and 580 and 680 °C, respectively. The declinations of both components are similar but in the sites beneath the Stac Fada Member (sites 3–10 in Figs 3 and 4) the magnetite-resident components tend to be variably shallower in inclination than the hematite components. This effect is seen, for example, in samples from sites 5, 6 and 7 illustrated in Figure 10, and this difference between magnetite and hematite-resident inclinations is highest at sites 5–8; it is not evident at sites 3 and 4. Group mean directions derived from magnetite and hematite components in sites below 8 are $D/I = 306/18^\circ$ ($\alpha_{95} < 7.1^\circ$) and $D/I = 290/31^\circ$ ($\alpha_{95} = 6.4^\circ$). Sites 9 and 10, an uncertain stratigraphic height above 1 to 8, show no clear distinction between the two components (Table 3).

Table 3. Overall group mean directions and pole positions for magnetite and hematite resident components in the Stoer Group succession

	T_B	N	R	k	α_{95}	D	I	Latitude	Longitude	dp/dm
Sites 5–8	Low	21	20	21	7.1	306	18	26.2	237.4	3.8/7.4
	High	12	11.8	47	6.4	290	31	24.7	256.5	4/7.2
	Total	33	31.4	19.9	5.7	300.4	23.1	26	244.3	3.2/6.1
Sites 9–10	Low	8	7.6	15.8	14.4	301	31	30.3	246.4	9/16.1
	High	8	7.9	55.5	7.5	314	35.8	39.2	235.4	5/8.7
	Total	16	15.4	23	7.9	307.5	33.5	34.9	241.1	5.1/9
Sites 14–21	Low	39	35.5	11	7.3	310.6	37	38.4	239.5	5/8.6
	High	38	33.5	8.1	8.7	306.6	40.4	38.5	245.1	6.4/10.5
	Total	77	68.9	9.4	5.6	308.7	38.6	38.4	242.1	4/6.7

D and I are the mean declination and inclination derived from N samples yielding a vector of resultant R and Fisher precision parameter k (= (N − 1)/(N − R)). α_{95} is the radius of the cone of confidence about the mean vector and dp and dm are the radii of the oval of 95 % confidence about the pole position in the colatitude direction and at right angles to it, respectively.

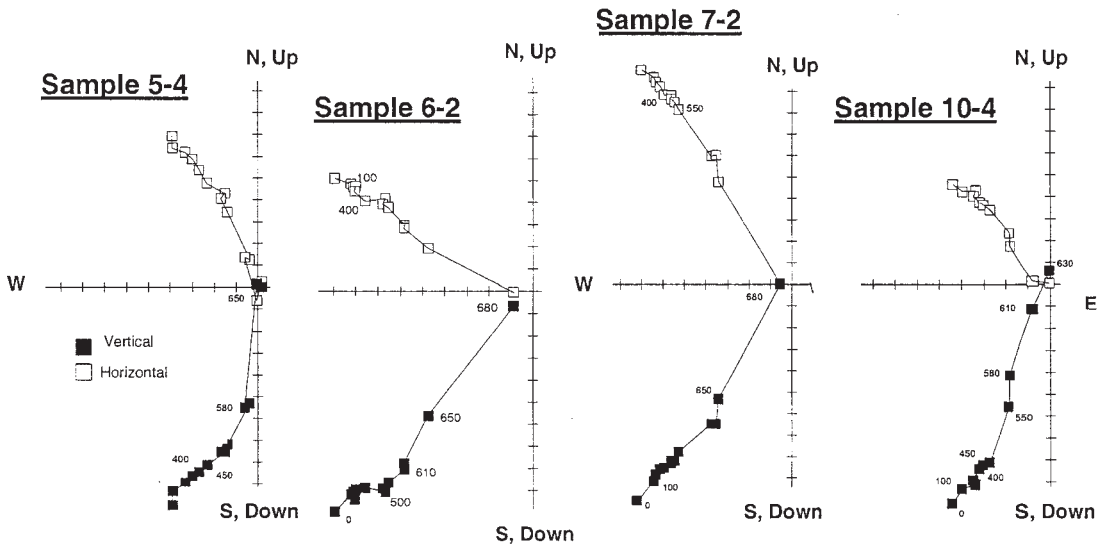


Figure 10. Orthogonal projections of typical thermal demagnetization results from Stoer Group sediments from levels below the Stac Fada Member. The magnetization vector at each stage of treatment is projected on to the horizontal and vertical planes as open and solid symbols respectively. The figures refer to demagnetization temperatures in °C.

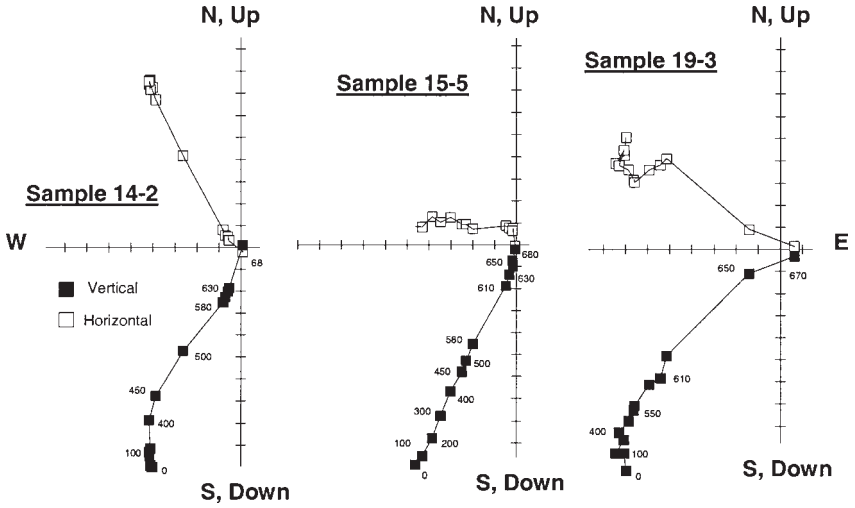


Figure 11. Orthogonal projections of typical thermal demagnetization results from Stoer Group sediments above the level of the Stac Fada Member. Symbols as in Figure 10.

At sites above the Stac Fada Member (sites 14–21 in Figs 3 and 4) some of the samples show up to three-component structures (e.g. 14-2 and 19-3 in Fig. 11), possibly due to higher percentages of fine pigmentary hematite (Section 4.a), although the lowest blocking temperature component is usually too short for accurate definition or is close to the present field direction. The distinction between shallower inclination magnetite and steeper inclination hematite components is less marked at these sites; the two may merge to define a single component, although a distinction is usually discernible (samples 14-2 and 15-5 in Fig. 11). Rarely the magnetite component has steeper inclination than the hematite component in this section (19-3 in Fig. 11). In this higher part of the succession, the overall distribution of magnetite components tends to be of somewhat shallower inclination than the hematite components, although both are fairly dispersed (Fig. 12). As a result, the means defined in the magnetite ($D/I = 310.6/36.9°$, $\alpha_{95} = 7.3°$) and hematite ($D/I = 306.6/40.4°$, $\alpha_{95} = 8.7°$) ranges are not significantly different, although the apparent contrast is in the same sense as that found in the lower part of the Stoer Group. We have found no evidence for a NE+ component, suggested by Torsvik & Sturt (1987) to be indicative of Mesozoic overprinting.

5.b. The Stac Fada Member

The magnetizations of clasts of Stoer Group sandstone incorporated into the Stac Fada Member were considered by Irving & Runcorn (1957) to show that sediments beneath this level had been magnetized by

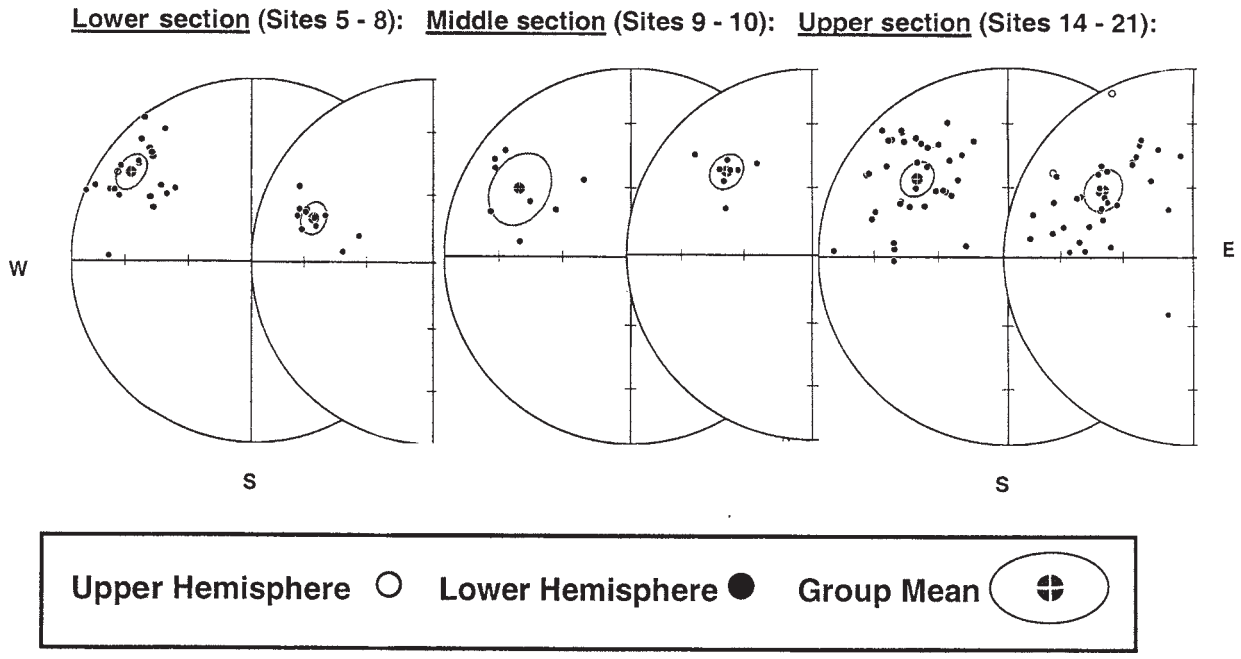


Figure 12. Equal Area projections showing directions of components of magnetization from groups of sites through the type Stoer Group succession at Stoer Bay before (left hand side) and after (right hand side) adjustment for tilt. The crossed circle symbol is the mean and the oval is the cone of 95 % confidence about the mean direction.

the time of emplacement of this volcanogenic horizon. Stewart & Irving (1974) noted contrasting remanence directions in the Stac Fada Member (D/I = 308°/37°) in the Stoer Peninsula and a direction of D/I = 283°/53° at Enard Bay isolated at unblocking temperatures of 500 °C to 600 °C. The present study has compared the palaeomagnetic record in the Stac Fada Member at sites 11–13 in the outcrop at Stoer, and 22 and 23 in the outcrop at Enard Bay.

These sites show one- or two-component behaviours in the magnetite range (Fig. 14) with this mineral identified by the rock magnetic experiments (Fig. 13). Component directions have similar declination but steeper inclination than directions observed in the sediments. At site 23 from Enard Bay a reversed direction is preserved in four cores in the hematite range (sample 23-3 in Fig. 14); this is the only evidence for field reversal so far discovered within the Stoer succession (Fig. 15). Samples from three Stoer sediment clasts incorporated into the Stac Fada Member (site 13) have accordant component directions and appear to have been remagnetized (Table 4). Low blocking temperature components from sites 11 and 12 are of steep inclination but distinct from the Earth's present field direction and of uncertain origin. Excluding these components, a tilt-adjusted mean direction of D/I = 304.6°/48.7° ($\alpha_{95} = 6.1°$) is derived. Although this is only slightly steeper than typical inclinations in the upper part of the Stoer Group, there is a distinct difference between the direction of magnetization in site 11 at the base of this unit and site 12 near the top (Table 4). A difference between

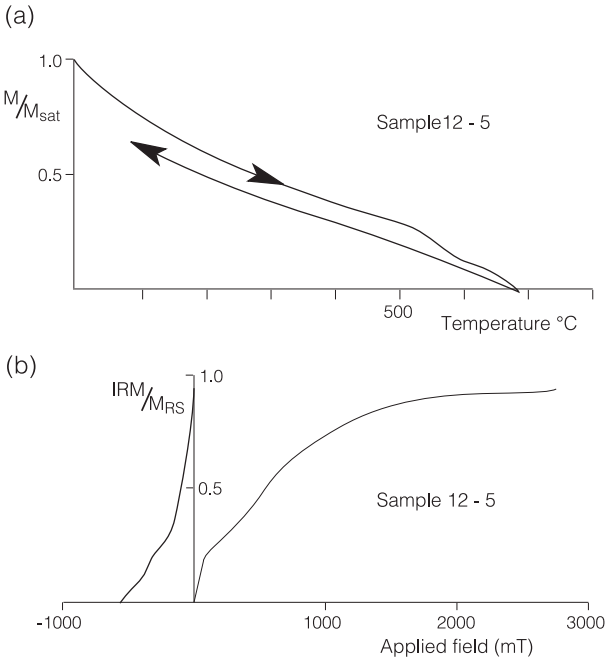


Figure 13. Examples of palaeomagnetic and rock magnetic results from the Stac Fada Member. (a) Thermomagnetic determination indicating magnetite (by a subdued inflection during heating) and hematite. (b) IRM determination; note the presence of both high and low coercivity constituents.

the lower and upper parts of this unit (although not significant at the 95 % confidence level in this case) in the same sense is also observed in the Stac Fada Member at Enard Bay and appears to account for the observation of Stewart & Irving (1974) noted above.

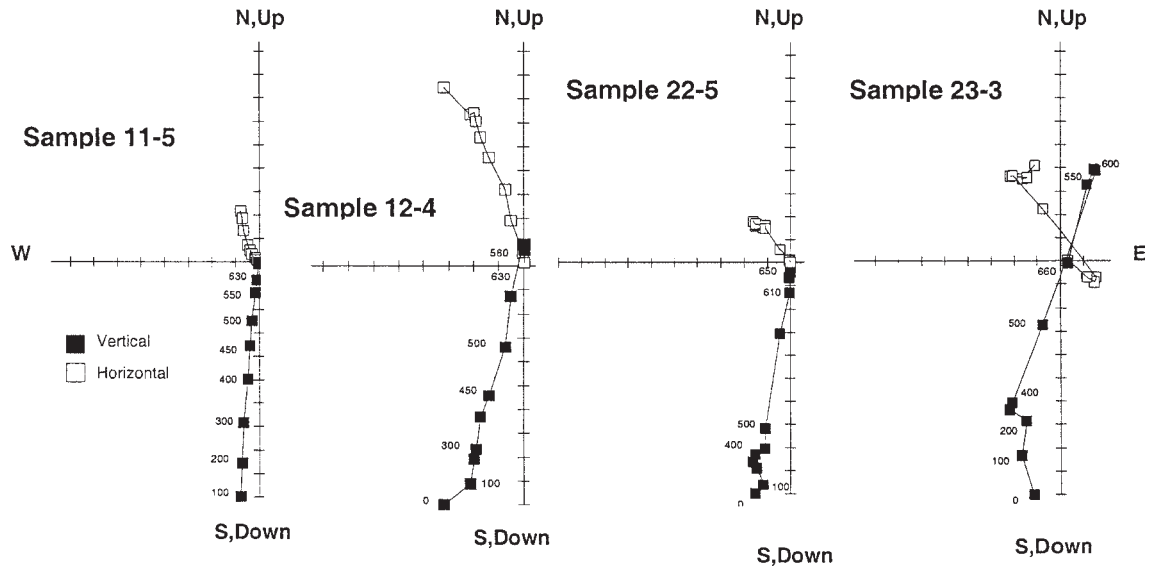


Figure 14. Orthogonal projections of the magnetic remanence during thermal demagnetization of samples from the Stac Fada Member; symbols as in Figure 10.

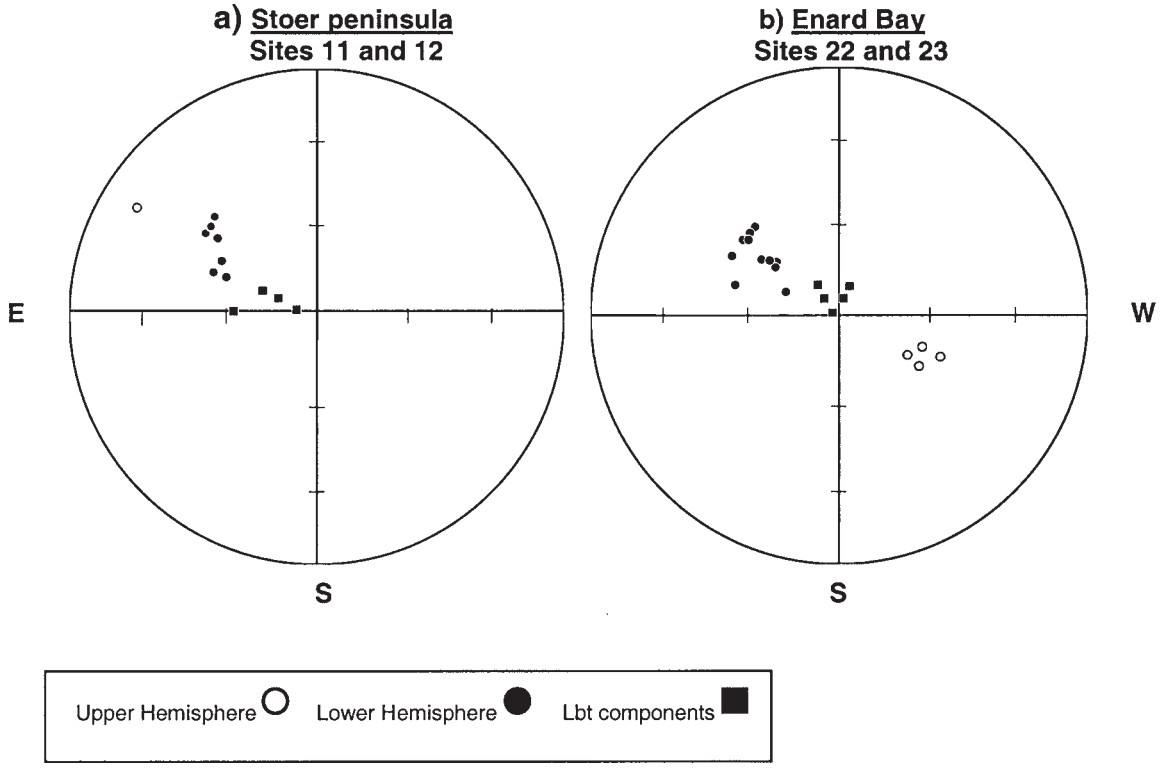


Figure 15. Component directions of magnetization observed in the Stac Fada member in outcrops at the Stoer Peninsula and Enard Bay after tilt adjustment. The squares refer to the components of low blocking temperature (Lbt) and the circles to components of high blocking temperature (see text). Closed symbols are lower hemisphere projections and the open symbols are upper hemisphere projections.

Possible causes for this must take into account the two explanations for this unit in the context of mudflow or peperite theories of origin (Sanders & Johnston, 1989; Stewart, 1990). Sanders & Johnston (1989) argue that the Stac Fada Member is a peperite, emplaced as fluidized slurry. They picture the unit as emplaced downwards into underlying sediment and forwards of an advancing mudflow with blocks of the underlying semi-consolidated sediment becoming dislodged and incorporated to form an angular breccia. The mixture of basic volcanic lapilli in this sediment is the signature of its volcanic origin (Lawson, 1972).

Table 4. Site mean palaeomagnetic results for sites in the Stoer Group

	In situ						Tilt adjusted	
Site	N	R	k	α_{95}	D	I	D	I
1+	28	15.3	2.1	25.6	326.3	59.3	302	36.2
2+	10	4.9	1.8	53.6	271.7	0	272	−30
3	8	7.6	17	13.9	349.6	66.6	309.1	51.7
4	12	11	11.2	13.5	323.2	39.5	311.4	20.9
5	10	9.1	10.5	15.6	308.8	50.1	297	27.2
6++	12	11.5	23.9	9.1	306.8	41.3	293.7	17.1
7	6	5.9	42	10.4	318.5	42.2	306.8	22
8++	6	5.7	16.4	17.1	320.6	42.4	308.2	22.8
9	8	7.8	30.8	10.1	320.1	44.2	307.1	24.4
10	8	7.8	30.3	10.2	335.2	60.1	308	42.6
11	7	6.9	43.2	9.3	17	80.7	291.7	65.2
12	4	3.99	542	4	337.6	60	310	44.1
13*	9	7.1	4.2	28.5	301	43	332	−16
14	26	23.9	11.8	8.6	318.7	51.4	304	33.1
15	14	9.75	3.1	27.5	348	60.3	315.6	47.4
16	7	6.7	23	12.8	317	49.6	303.2	30.2
17	8	7.4	11	17.5	324.8	61.3	310	41.1
18	8	7.3	10.2	18.2	314.1	47.1	303.7	31.4
19	9	8.6	20	11.8	309.3	55.5	297.3	38.6
20	10	9.1	10.5	15.7	17.4	52.5	341.3	52.6
21	7	6.4	10.1	20	315.1	47.4	304.3	32.2
22	7	6.9	77.6	6.9	302	74.6	297	62.7
23	8	7.9	98.1	5.6	309.6	66.2	304.4	54.3
24	5	4.9	62.6	9.7	316.8	29.7	303	35.8

N = number of components. *assumes the magnetic component is a Stoer overprint. + conglomerate test; ++ slump tests. Other symbols as in Table 3.

Stewart (1990) rejects the peperite theory of Sanders & Johnston (1990) and notes that the Stac Fada Member is mostly conformable with the underlying sediment. He considers that the unit underwent a post-depositional mass movement down a regional palaeoslope, which was locally responsible for producing the breccias. With the peperite model, the primary magnetization in the Stac Fada Member is likely to have been depositional to post-depositional as magnetic grains rapidly settled or became oriented during consolidation. In the model invoking a muddy sandstone deposit subject to slippage, it is possible that the remanence could be a shear remanent magnetization (SRM; see Games, 1980) acquired over a short period of time and therefore recording a virtual geomagnetic field direction.

The above discussion does not explain the difference of magnetic inclination between the lower and upper parts of the Stac Fada Member. However, it could be that a chemical process magnetized the upper part, presumably exposed for a period at the surface, sometime after the lower part had consolidated following acquisition of a short-term magnetization (yielding a virtual geomagnetic pole). The upper part (which is stratified above a lower massive unit) could therefore record a younger and time-averaged magnetization equivalent to a palaeomagnetic pole. Certainly the direction recorded in this top part of the unit is not inconsistent with directions of magnetization resolved in the sediments within the upper part of the Stoer Group (sites 14–21, Table 4).

5.c. Palaeomagnetic field tests

5.c.1. Slump tests

Sites 6 and 8 are sampled in two horizons which evidently slumped when the sediment was still a waterlogged slurry and presumably soon after deposition. Comparable slumps in the Stoer Group are described by a number of authors and may have been triggered by earthquakes (Selley, 1969). Component behaviours are comparable to undeformed sediments above and below these horizons with a shallower inclination magnetite component and a steeper inclination hematite component (Fig. 16a). When adjusted for tilt of the bedding at the locations of the cores, the scatter of component directions is increased (Fig. 16b) and the ratio of precision parameters (k_1/k_2) is 2.4 at site 6 and 1.5 at site 8 (Table 5). The former yields a negative fold test at the 95 % confidence level (McElhinny, 1964) indicating a post-slumping remanence. These tests show that the remanence in the Stoer group sediments cannot be a pure depositional remanent magnetization (DRM).

5.c.2. Stoer Blocks in the basal Torridon Group

Prior to deposition of the overlying Torridon Group at ~1030 Ma, the sub-Torridon unconformity near Achiltibuie on the north side of Loch Broom comprised an irregular topography that included upstanding relicts

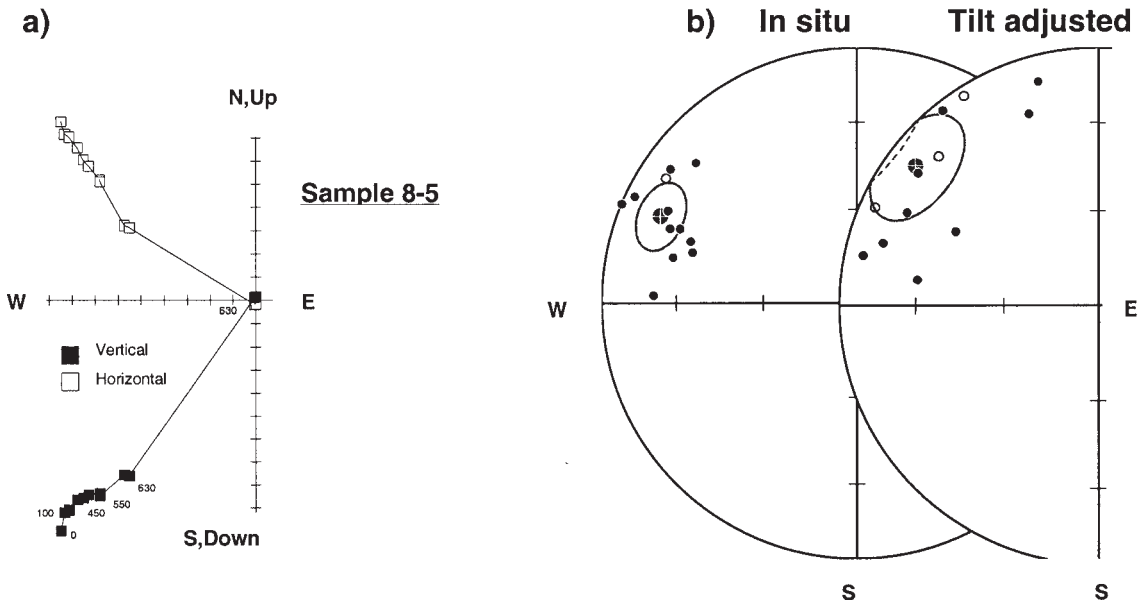


Figure 16. Palaeomagnetic results from soft sediment slump structures at sites 6 and 8. (a) Typical orthogonal projection of a sample demagnetization; symbols as in Figure 10. (b) Component directions of magnetization in samples distributed across a sedimentary slump at site 6 before and after correction for local tilt of the bedding at the core location. The mean directions are shown (crossed larger circle) with circles of 95 % confidence; closed symbols are lower hemisphere and open symbols are upper hemisphere plots.

Table 5. Palaeomagnetic results from sedimentary slumps in the Stoer Group

	In situ					Tilt adjusted					
Unit	D°	I°	$\alpha_{95°}$	N (R)	k_1	D°	I°	$\alpha_{95°}$	N (R)	k_2	k_1/k_2
Site 6	294	17	9	12 (11.5)	24	307	12	15	12 (10.8)	10	2.4
Site 8	308	23	17	6 (5.7)	16	309	17	21	6 (5.6)	11	1.5

Symbols as in Table 3.

of the Stoer Group. Blocks from cliffs in this group broke away and slipped into the Diabaig Formation as deposition of the Torridon Group progressively buried the underlying topography (Stewart *in* Barber *et al*. 1978). Six of these blocks ranging from 1 to 5 metres in size were sampled and cores were also taken from the host Diabaig Formation. Way-up and bedding orientations were recorded from the blocks in each case.

Magnetizations show a range of two- and three-component structures, but as with *in situ* samples from the Stoer Group elsewhere, distinct components are resolved up to 580 °C and between 580° and up to 675 °C (Fig. 17a). The former tend to be somewhat variable in direction even within the same block but the latter are always coherent within blocks and are used for interpretation here. Two contrasting groups of directions of magnetizations, with each group composed of three clasts, are present. One group has a steep E+ direction and a shallow W+ direction (Fig. 17b). The second direction is unlike any remanence observed in the host Torridon Group. Although the Diabaig Formation at this location (where the tilt is shallow to the south) has an E+ remanence (sample 27-7 in Fig. 17 and Table 6) comparable to the 'B' overprint of Caledonian age extensively recognized in the Torridon Group (Darabi & Piper, unpub. data), the ChRM directions in the clasts are mostly steeper or have contrasting declinations and are clearly different from the host sediment (Fig. 17, Table 6). Torsvik & Sturt (1987), who sampled some of the blocks used for this study, resolved similar contrasting sets of directions.

When the bedding within the blocks is restored to the horizontal to perform an inclination-only test, the inclinations approach an intermediate positive value ($I = 32°$) close to the mean inclination in the Stoer Group ($I = 36°$). The group mean inclination changes from $I = 45.8°$ ($\alpha_{95} = 25.9°$) *in situ* to $I = 32.3°$ ($\alpha_{95} = 7.3°$), an improvement which is significant at the 95 % confidence level according to the inclination-only test of Kono (1980). The results of this test (Table 6) are therefore similar to the test performed by Torsvik & Sturt (1987) and confirm that the magnetic remanence was already fixed into the Stoer Group before deposition of the Torridon Group at ~ 1030 Ma.

5.c.3. Conglomerates and breccias in the Stoer Group

Boulder beds within the Stoer Group were palaeomagnetically investigated by Piper & Poppleton (1991), who found that the clast magnetizations had

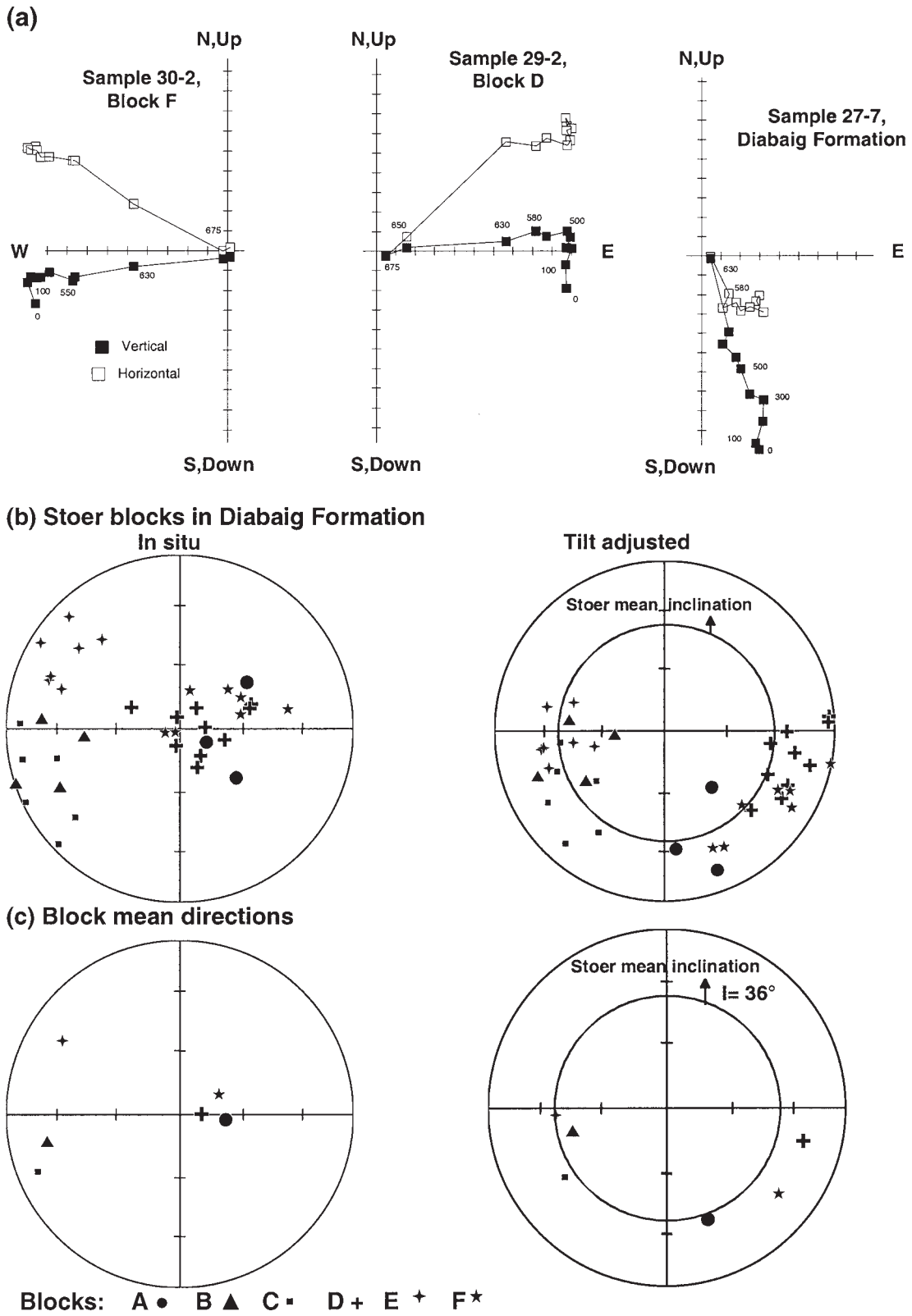


Figure 17. Palaeomagnetic results from blocks broken off a palaeocliff in the Stoer Group and slumped into the Diabaig Formation of the basal Torridon Group. (a) Orthogonal projections of two samples from Stoer group blocks and a sample of host Diabaig Formation sandstone; symbols as in Figure 10. (b) Directions of high blocking temperature components of magnetization before and after adjustment for tilt of the bedding in the clasts. (c) Block mean directions of magnetization. The inner circles with an inclination of 36° in (b) and (c) define the mean tilt-adjusted inclination of the magnetization in the Stoer Group elsewhere.

Table 6. Summary of palaeomagnetic results from slumped boulders in basal Diabaig Formation at Achiltibuie

Block	Field numbers	No. of cores	Dip direction	*In situ*					Tilt adjusted	
				D°	I°	α_{95}°	N (R)	k	D°	I°
A	27-1 to 4	4	54/190	97	65	39	3 (2.8)	11	160	34
B	28-1 to 4	4	22/85	259	23	25	4 (3.8)	14	257	45
C	28-5 to 7	3	53/72	248	13	17	6 (5.7)	16	236	32
D	29-1 to 4	4	34/119	89	80	12	10 (9.5)	17	104	23
E	29-5 to 8	4	55/103	302	22	11	7 (6.8)	30	267	38
F	30-1 to 4	4	58/180	62	69	17	7 (6.6)	13	127	23
Diabaig Fm.	27-5 to 10	6	18/172*	91	74	19	8 (7.3)	10	91	74

*Regional dip of Torridon Group sediments at this location. Symbols as in Table 3.

generally survived erosion and deposition. Although there was local evidence for Mesozoic remagnetization, no systematic overprinting was recognized. Two further horizons have been investigated in this study at exposures on the north side of the Stoer Peninsula (Fig. 3). Site 2 is at the unconformity where locally dislodged blocks of Lewisian amphibolites occur in a sandy matrix and site 1 is sited in a boulder bed separated from the unconformity by bedded sandstones. At site 1, magnetizations in the amphibolites include components subtracted above 580° (Fig. 18a), and although directions tend to be west of north and most commonly of positive inclination, no systematic overprint is recognizable (Fig. 18b). It is possible that a few amphibolite clasts carry a Stoer overprint (sample 1–3 in Fig. 18a), but there is no sign of a Torridon Group primary or secondary remanence.

Amphibolites from the unconformity (site 2) have only poorly stable magnetizations. This is probably a result of weathering because rock magnetic tests show a dominant paramagnetism. The *in situ* directions from amphibolites include components with westerly positive directions (crosses in Fig. 18c), similar to the A2 direction widely recorded in the underlying Scourian (Lewisian) basement (Piper, 1992). In contrast, dislodged clasts have apparently random directions not obviously interpretable in the context of later geological events in this region and specifically with overprints in the overlying Torridon Group.

5.c.4. Fold test

The palaeomagnetic results from the Stoer Group come from two locations with the structural tilts across each outcrop almost constant. The scope for a palaeomagnetic fold test is therefore limited and is dependent on the contrast between steeper dips to the west in the Stoer Peninsula and the very shallow dip to the west at Enard Bay. We exclude results from the Stac Fada Member from this analysis because of the contrast in magnetic inclinations within this unit and the possibility that at least part of this is the record of a virtual geomagnetic field direction and not a time-averaged palaeomagnetic direction (Section 5.b). When tilt adjustment is applied there is an apparent improvement in the overall grouping of inclinations and a convergence is also evident in magnetic declinations (Fig. 19). The statistics are summarized in Table 7 and indicate that the Stoer Group probably preserves a pre-folding magnetization although the test is not significant at the 95 % confidence level.

Table 7. Mean direction of all sites before and after tilt adjustment

		N	R	k	α_{95}	D	I
Site level	*In situ*	24	21.9	11.2	9.3	320.5	51
	Tilt adjusted	24	22.5	15.7	7.7	303.9	36.2
Sample level	*In situ*	242	187.2	4.4	4.9	320.2	52.3
	Tilt adjusted	242	186.4	4.3	4.9	304.4	35

Symbols as in Table 3.

6. Discussion and regional interpretation

The rock magnetic and palaeomagnetic evidence shows that magnetite and hematite are both present as ferromagnetic constituents in the Stoer Group sediments and both carry ancient components of magnetization. The magnetite is most likely to be detrital and derived from erosion of the local Scourian basement (Young, 1999; Rainbird, Hamilton & Young, 2001). The relative significance of detrital versus authigenic hematite is unclear, but hematite staining is ubiquitous and occurs around specularite and silicate grains and as interstitial cement (Stewart & Irving, 1974). The most significant features of the simple magnetic component structure of most of the Stoer Group samples are the similarity of the magnetite and hematite-resident component directions and the widespread tendency for the magnetite components to have somewhat shallower inclinations than the hematite components. This is most logically interpreted as due to (1) partial oxidation of detrital magnetite grains to hematite during diagenesis; (2) rotation of differing size fractions of these heavy minerals during post-depositional processes; and/or (3) precipitation of the hematite as a grain coating or as authigenic grains during later stages of diagenesis.

An early origin for the NW+ remanence in the Stoer Group was indicated by observation that Stoer sediments disturbed by emplacement of the Stac Fada

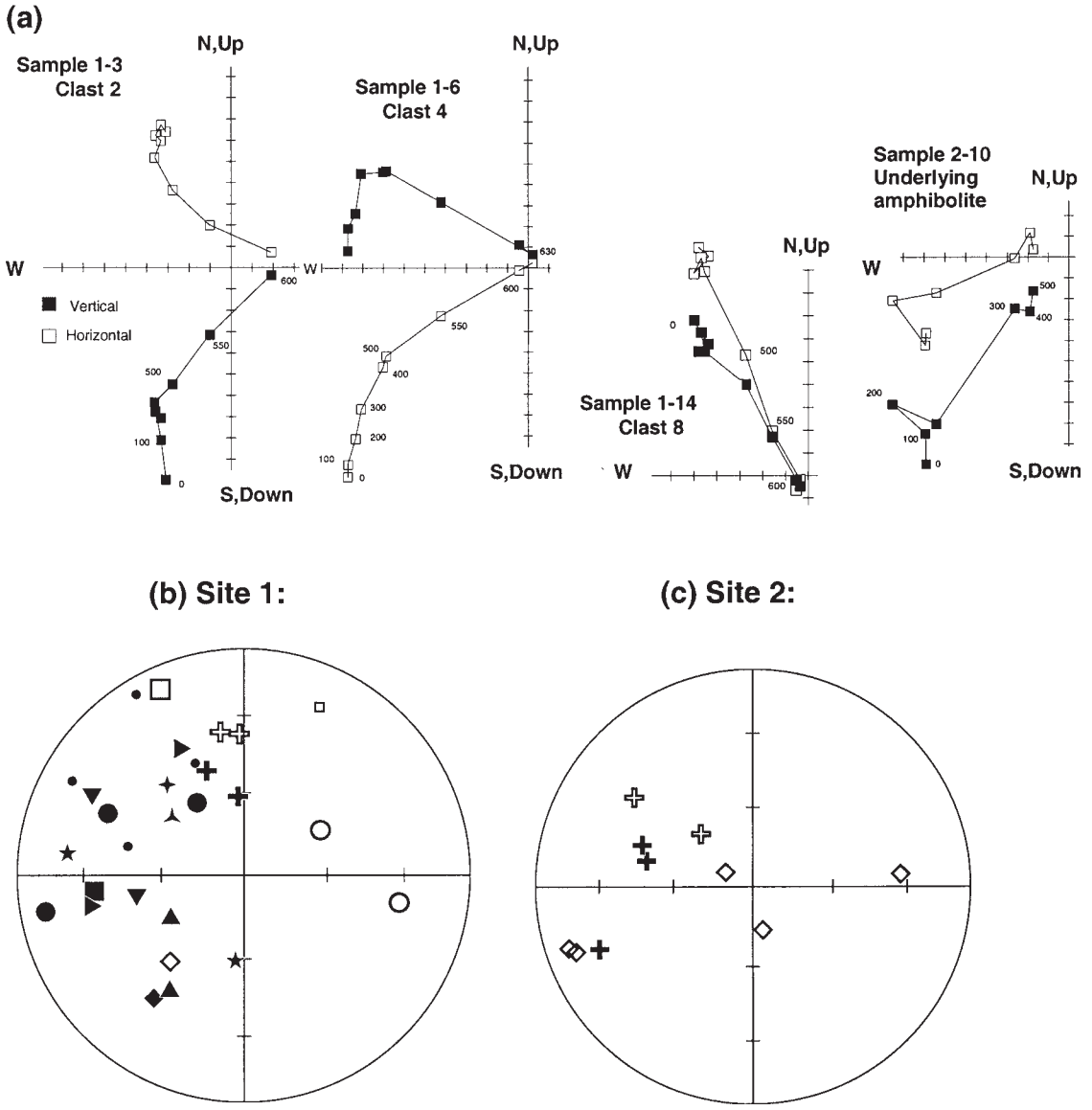


Figure 18. Palaeomagnetic results from amphibolite clasts at, or close to, the unconformity between the Stoer Group and the Lewisian metamorphic complex. (a) Orthogonal projections showing thermal demagnetization results from clasts at sites 1 and 2; symbols as in Figure 10. (b) Component directions of magnetization from Lewisian clasts in the boulder bed at site 1; symbols refer to results from different clasts with closed symbols being lower hemisphere plots and open symbols being upper hemisphere plots. (c) Component directions of magnetization from amphibolite in clasts above the unconformity (diamonds) and underlying Lewisian basement (crosses).

Member appeared to show random magnetizations (Irving & Runcorn, 1957; Stewart & Irving, 1974). A Detrital Remanent Magnetization (DRM) caused by alignment of magnetic grains with the external field during deposition cannot, however, be applicable to the Stoer Group because slumps formed whilst the sediment was still wet slurry have uniform magnetizations. Furthermore, the acquisition of DRM by specularite tends to be reflected in poor directional grouping due to weak magnetic torques and hydrodynamic randomizing forces. A Post-depositional Detrital Remanent Magnetization (PDRM) of early diagenetic origin, presumably facilitated by compaction and dewatering, is therefore indicated. Furthermore, the systematic relationship between k_{max} and remanence, which has been taken to indicate a DRM origin in hematite (Løvlie, 1974), is not apparent in these sediments (Figs 7, 12). The enhancement of grain orientation to impart a PDRM is likely to overlap with chemical change responsible for a chemical remanent magnetization. In red sediments this can occur through the oxidation of detrital magnetite, alteration of iron-bearing silicates, and through pigmentation. The grains of detrital and/or chemical (authigenic) origin in these sediments are evidently the main remanence carriers, because acid leaching of the pigment indicates that it makes no significant contribution to the total NRM (Smith, Stearn & Piper, 1983), presumably because the domain structure is unsuitable. Since the single domain–multi-domain transition is ~ 0.1 mm (Kletetschka &

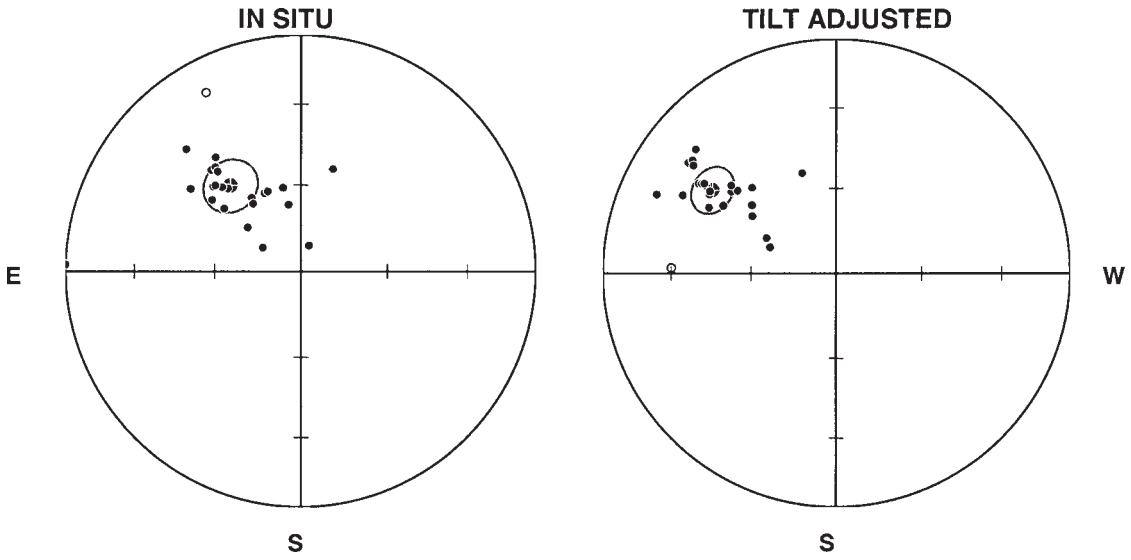


Figure 19. Equal area projection showing site mean directions of magnetization in the sedimentary sites from the Stoer Group before and after adjustment for tilt. The overall group mean directions and circles of 95 % confidence are also shown: closed symbols are lower hemisphere and open symbols are upper hemisphere plots.

Wasilewski, 2002), most larger grains sizes could potentially contribute to the remanence.

The Stoer Group is unique in the sedimentary cover on the northwest foreland of Scotland in preserving a record of Proterozoic deformation. The later history recorded by the Torridon Group and the Early Palaeozoic succession is dominated by syndepositional extension. The Stoer Group also has a quite different history of magnetization from the Torridon Group. Younger overprints are virtually absent in the former group. In contrast, the Torridon Group has a complex magnetization with secondary overprinting of Caledonian and younger age partially, and sometimes completely, replacing a primary dual polarity NW−/SE+ remanence of PDRM origin (Darabi & Piper, unpub. data). As shown by the results from the slumped blocks in basal Torridon Group sediments at Achiltibuie, lithification of the Stoer Group was complete by the time that the Torridon Group was deposited at ~1030 Ma. In contrast, removal of porosity in the Torridon Group was still incomplete during the Caledonian Orogeny in mid-Ordovician times when widespread authigenesis, probably by forced fluid flow, imparted a secondary overprint (Darabi & Piper, unpub. data). Secondary overprints continued to be locally acquired by the Torridon Group during Mesozoic rifting of the Minches Basin. We have found no evidence for such an overprint in this study of the Stoer Group, although Torsvik & Sturt (1987) report traces of a Mesozoic remanence, and an overprinted remanence of this age was also detected in a Stoer breccia by Piper & Poppleton (1991).

Although the origin of the range of magnetic inclinations resulting from earlier studies of the Stoer Group was unclear (Table 1), it is now apparent that it records a sequential change in the direction of the magnetizing field during the interval of deposition and diagenesis. A time sequence of poles or apparent polar wander path (APWP) is therefore defined by the stratigraphic succession of poles. The consistent sense of motion represented from shallower to steeper inclination is recorded whether the low and high blocking temperature components are considered separately or together (Table 3). Poles of comparable age to the 1199 ± 70 Ma Pb–Pb determination from the Stoer Group (Turnbull, Whitehouse & Moorbath, 1996) are unrepresented in the North American sector of the Laurentian Shield where there is a time gap in the record between the Mackenzie igneous episode at ~1250 Ma and the oldest part of the Keweenawan track at ~1110 Ma. The intervening period is, however, covered by the record from the Gardar Igneous Province of South Greenland (Piper, 1977; Piper *et al*. 1999). The relevant time sequence of poles from Greenland with assigned age dates is summarized in Table 8.

When the Stoer Group poles are rotated into the Greenland reference frame (22° clockwise about a Euler pole at 73.0°N, 96.5°E) to the pre-drift configuration of the North Atlantic continents (Bullard, Everett & Smith, 1965), the sequence of poles from lower to higher beds in the Stoer group (Table 3) is rotated to 221°E, 22°N; 217°E, 30°N and 218°E, 34°N. This sequence of poles conforms to the sense of motion along the Gardar Track and plots between groups of poles with ages averaging 1204 and 1158 Ma on the Laurentian reference path (Fig. 20). Precise correlation is observed (Table 8) with the pole from the Hviddal Giant Dyke (1180 ± 9 Ma), but the Stoer poles are clearly older than the Younger Giant dykes (1167 ± 3 Ma). The Pb–Pb age determination from the Stoer Group (Turnbull, Whitehouse & Moorbath, 1996) is therefore supported as a depositional/early diagenetic age.

Table 8. Summary of palaeomagnetic results defining the Gardar palaeomagnetic track

	Rock unit	Radiometric age (Ma)	Geological age	D/I	Palaeomagnetic pole °E	°N	dp/dm	λ	P
1a	Eriksfjord Group basal Lavas and Carbonatite		> 1b	290/1	207	10	7/14	0	M
1b	Eriksfjord Group, Lower Lavas		> 1c	260/25	239	7	7/14	13	M
1c	Eriksfjord Lavas, Upper Lavas		> 4	280/24	222	16	5/9	13	M
2	Grønnedal-Ika Complex	1299 ± 17		271/21	228	9	3/5	11	R
3	Motzfeldt Complex	1350 ± 10	> 14	286/4	212	10	6/12	2	R
4	North Qôroq Complex	1291 ± 61	> 14	297/2	201	14	3/6	1	R
5	Narssassuaq Stock			288/1	207	9	4/7	0	R
6	Lamprophyre Dykes			285/−9	206	3	5/10	4	R
7	BD0 Dyke	1284 ± 3, 1275 ± 3	> 9	264/−19	203	−12	3/5	10	N
8	Kûngnât Fjeld Complex	1286 ± 2		292/−15	199	3	2/4	8	R
9a	BD1 Dykes, western sector		< 7	284/−7	308	4	8/16	3	R
9b	BD1 Dykes, north-west sector		< 7	277/−1	215	3	5/10	0	R
9c	Dolerite dykes cutting Grønnedal Complex		< 2	275/12	220	7	4/9	6	R
10a	BD2-BD3 Dykes, western sector		< 9	287/43	220	30	5/7	25	R
10b	BD2-BD3 Dykes, NW sector		< 9	288/56	227	40	9/13	37	R
10c	BD2-BD3 overprints in Unit 1		< 9	267/48	229	25	6/9	29	R
11	Hviddal Giant Dyke, Tugtutôq	1180 ± 9		294/43	215	33	8/12	25	R
12	Narssaq Gabbro and Ulramafics			285/47	225	32	8/12	28	R
13	Younger Giant Dykes, Tugtutôq	1167 ± 3, 1163 ± 2	< 11	292/57	226	42	8/11	37	R
14	South Qôroq Complex	1160 ± 8		301/51	214	42	5/7	32	R
15a	BD3 Dyke swarms, eastern sector			290/51	224	36	4/6	32	R
15b	BD3 Dyke swarms, NW sector			283/51	231	33	5/7	32	R
16	Eriksfjord Redbeds, B overprint			284/67	244	48	8/9	50	R
17	Steep Inclination Dykes, W sector			262/75	268	48	22/24	62	R
18	Ilímaussaq Complex	1145 ± 5	< 15	327/81	282	73	12/12	72	M

D/I are the mean declination/inclination of the palaeomagnetic study, dp and dm are the radii of the oval of confidence in degrees about the pole position in the colatitude direction and at right angles to it respectively. λ = palaeolatitude; polarity is given as R = reversed, N = normal or M = mixed. Pole summary after Piper (1977) and Piper *et al.* (1999). In the column for geological age the numbers refer to the relative age of the units established by stratigraphic relationships.

Field and geochronological studies during the past two decades have revealed a protracted and complex history for Grenville terranes along the Laurentian margin with episodes of deformation and metamorphism separated by periods of extension recorded by rifting and clastic sedimentation. In Labrador, for example, the principal phase of high-grade deformation occurred between 1080 and 1050 Ma, although anorogenic magmatic activity extended back at least 400 Ma before this (Corrigan, Rovers & Dunning, 2000). Whilst culminating phases of the Grenville Orogeny including high grade metamorphism are recognized in most sectors of the belt at ∼1050–950 Ma (Scharer & Gower, 1988; Gower *et al*. 1991; Corrigan, Rovers & Dunning, 2000), arc accretion was a key part of the prior evolution of the belt and was interspersed with basin development at ∼1280–1250 and 1180–1150 Ma in the Ontario–Quebec and Adirondack segments (Hanmer *et al*. 2000). The latter interval correlates with Stoer Group sedimentation.

A more limited but still long (∼200 Ma) range of magmatic activity is recognized along the western extension of the Grenville Belt. A two-stage history is present in eastern Mexico and the Elzevirian arcs in Texas with magmatism and ductile deformation at ∼1250–1150 Ma, followed by granulite facies metamorphism at ∼1020–980 Ma (e.g. Lawlor *et al*. 1999; Lopez, Cameron & Jones, 2001; Weber & Kohler, 1999). In East Greenland, at least two phases of orogenesis are present with magmatism and intrusion concentrated at 950–920 Ma following deposition of the Krummedal supracrustal succession prior to 1000 Ma (Watt & Thrane, 2001), with the latter having its source in the collapse of an earlier orogen sometime between 1400 and 1100 Ma.

A degree of diachronism is almost certainly present along this extensive Grenville margin to the Laurentian Shield (Fig. 1) with Stoer Group deposition occurring at ∼1180 Ma and compatible with a short episode of extension before the culminating deformation and high grade metamorphism in the adjoining belt prior to deposition of the Torridon Group. The palaeomagnetic estimate for the post-depositional 'A' remanence in the Torridon Group is ∼1030–1020 Ma (Darabi & Piper, unpub. data) and subsequently this group was not significantly deformed until the Caledonian Orogeny. It could therefore have been a molasse deposited during uplift of the Grenville Orogen but was clearly deposited later than all major deformation in this sector. This is consistent with the history of eastern Labrador, for example (Scharer & Gower, 1988; Corrigan, Rovers & Dunning, 2000), although East Greenland was still undergoing high-grade deformation some 70 Ma later (Kalsbeek *et al*. 2000).

Palaeomagnetic comparisons identify a break-up and rotation of the Fennoscandian Shield with respect to Laurentia at ∼1000 Ma (Patchett, Bylund & Upton, 1978; Piper, 1980; Stearn & Piper, 1984). This event has been correlated with Torridon Group rifting and sedimentation culminating in the transition from

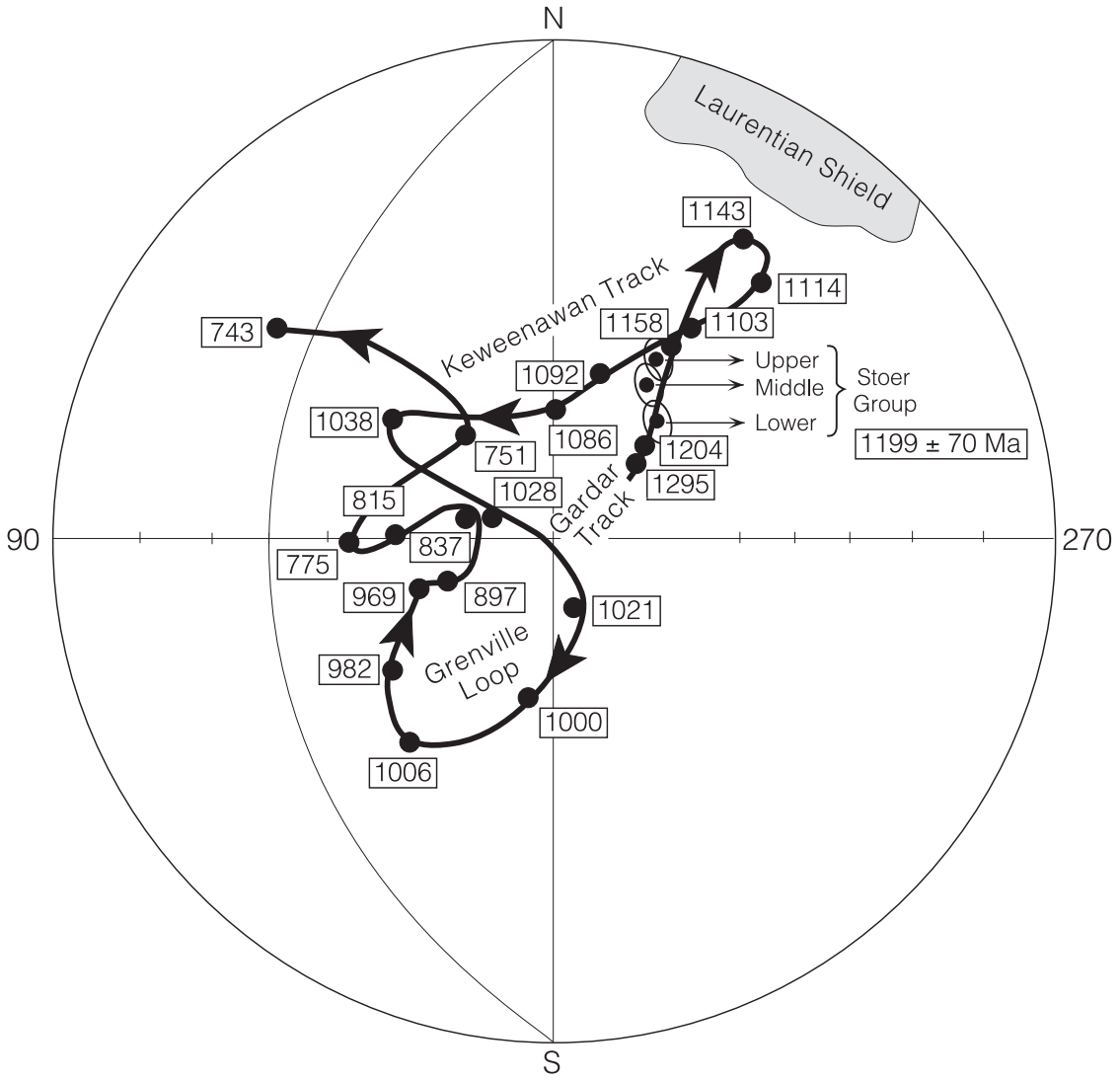


Figure 20. Pole positions and 95 % confidence ovals from the Stoer Group rotated into the Laurentian reference frame and compared with the ~ 1300–750 Ma apparent polar wander path from Laurentia. The mean poles with average ages for Laurentia are from Piper (2000).

terrestrial to marine sedimentation at the level of the Aultbea Formation (Piper, 1987). Subsequent deformation in East Greenland may correspond to a collisional event (Kalsbeek *et al*. 2000), as may post-kinematic magmatism at ~ 920 Ma in the Sveconorwegian Orogen at the Baltica margin (e.g. Eliasson & Schoberg, 1991). Since these events followed continental rifting and break-up and are confined to the sector east of the former site of the Hebridean Craton, they are probably to be interpreted as part of a post-Grenville history.

The possibility of a correlation between the Torridonian Supergroup on the Hebridean foreland and the Moine Supergroup within the adjoining Northern Highlands of the Caledonian orogenic belt has been much debated since it was first suggested by Geikie (1893). The apparent recognition of a Morarian Orogeny at ~ 750 Ma in the Grampian Group (Brook, Brewer & Powell, 1976; Piasecki & van Breemen, 1983), suggesting that the Moinian was appreciably older than this event, seemed to reinforce this view. However, the case against a Morarian Orogeny, and hence the absence of pre-Caledonian deformation in the Torridon Group, comprises (1) the absence of an orogenic unconformity within the Dalradian Group and (2) the recognition that the D2 deformation is Caledonian in age (Soper & England, 1995). Recent geochronological work has nevertheless reinforced the finding that a melting event occurred in the Northern Highlands at ~ 770 Ma (Friend *et al*. 1997; Kinny *et al*. 1999; Rogers *et al*. 2001). Whether this melting event was due to orogeny or crustal extension is not clear, although a correlation with basic magmatism (Miller, 1999) suggests the latter. The strong lithological case for a correlation between the Moine Supergroup and Torridon Group therefore remains. The Torridon Group has a substantial magnetic overprint that is apparently of Caledonian age (Darabi & Piper, unpub.

data), but the Stoer Group appears to have escaped the influence of any event later than the Grenville Orogeny.

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